

Workers' Voice in Westminster, 1832–2014

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London School of Economics

PSPE WiP Seminar

October 7, 2026

Motivation I

Early Brief

**Working-class Americans
aren't running for office. It's
changing our politics.**

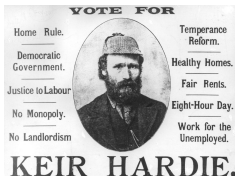
September 14, 2026

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Keir Hardie

Miner → MP, UK 1892

Lula: the rise and fall and rise again of Brazil's president-elect

Former shoe-shine boy who became country's first working-class president in 2002 sealed an astonishing comeback on Sunday

● [Live results page](#)



© Luiz Inácio Lula da Silva celebrates after winning Sunday's presidential run-off election in Brazil. Photograph: Nelson Almeida/AP/Getty Images

Lula

Metalworker → President, Brazil

Angela Rayner: Labour's working-class warrior who fell from power



Angela Rayner

Care worker → Deputy PM, UK

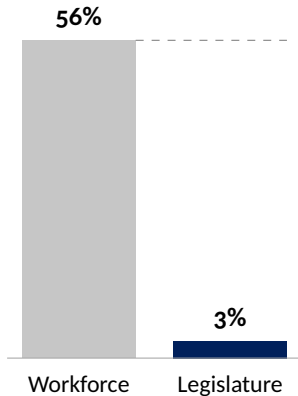


A. Ocasio-Cortez

Bartender → Congress, US

Motivation II

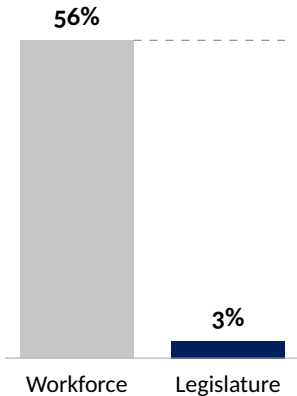
Median OECD country



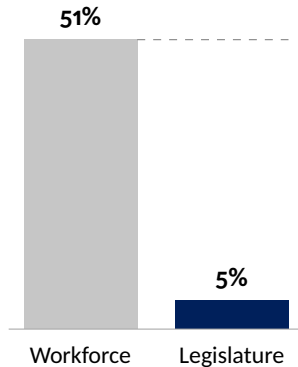
Notes: Share of working-class people in the labour force and in the national legislature. Source: Carnes and Lupu (2023b).

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Motivation III

- **Unequal responsiveness:** policy tracks the affluent rather than the median voter

(Gilens, 2012; Lupu and Warner, 2022; Lupu and Pontusson, 2023; Elsässer and Schäfer, 2023)

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- ▶ **Unequal presence:** workers are most of the workforce, but a small minority of legislators

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- ▶ *Risk aversion and resources.* Resources and gatekeepers screen workers out

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- ▶ *Benefits.* Selecting them could signal specific expertise (Costa and Pereira, 2026), attenuate their disadvantages (Abbott and DeVeaux, 2026) and mobilize voters (Poertner, 2023; Piketty and Cagé, 2023; Abbott and DeVeaux, 2024)

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Citizens $\rightarrow$ Aspirants $\rightarrow$ Candidates $\rightarrow$ **MPs** $\rightarrow$ **Office-holders**

The broader puzzle

- ▶ Canonical models: extend the franchise $\Rightarrow$ the decisive voter moves down the income distribution $\Rightarrow$ policy follows (Meltzer and Richard, 1981; Acemoglu and Robinson, 2006)

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First margin — *the vote*

Who gets to choose the legislator?

Extensively studied: 1832, 1867, 1884, 1918...

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First margin — *the vote*

Who gets to choose the legislator?

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Second margin — *the legislator*

Who actually sits in the chamber?

Persistently skewed toward the educated and the well-off

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- ▶ **Who serves:** rises before full suffrage, peaks post-war, declines after 1980; almost all Labour

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- ▶ **Legislator, not constituency preferences:** gap survives seat-fixed switching & a close-election RDD
- ▶ **Career advancement:** less likely to hold office within party, a gap largely explained by credentials

Contribution

- ▶ Elite persistence and descriptive representation
de jure vs *de facto* influence → bottom-up approach

(Acemoglu and Robinson, 2008; Pande, 2003; Chattopadhyay and Duflo, 2004; Fujiwara, 2015; Besley et al., 2023; Praninskas and Munroe, 2025; Langsæther and Nyrup, 2026)

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- ▶ Working-class representation.
Organized labour and the consequences of class gaps in legislatures

(Korpi, 1983; Przeworski and Sprague, 1986; Sassoon, 1996; Carnes, 2012; Wauters, 2012; O'Grady, 2019; Barnes and Saxton, 2019; Carnes and Lupu, 2023b; Fourniaies, 2024; Mor and Boix, 2024; Abbott and DeVeaux, 2026)

Setting and Data

UK Parliament, 1832–2014

Franchise and entry barriers

- ▶ **1832** Reform Act: electorate 0.4m → 0.65m; seats to industrial cities, labouring class still excluded
- ▶ **1867/1884** Urban artisans, rural labourers; first Lib-Lab MPs from the 1870s
- ▶ **1911** £400 MP salary removes the financial barrier to candidacy
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The Unions and the Labour Party

- ▶ **1900** Labour Representation Committee
- ▶ **1929** About half of Labour candidates union-sponsored
- ▶ **1945** Largest WC cohort; welfare state; union density near 55%
- ▶ **1979–** Deindustrialisation, pit closures; density below 30% by 2000
- ▶ **1996** Winnick amendment ends direct union sponsorship

Data: MP panel and speeches

MP panel, 1832–2014

- ▶ *Members after 1832* (History of Parliament Trust): biography, occupation, schooling, union links, offices held
- ▶ Enriched with legislatoR (Göbel and Munzert, 2022)
- ▶ *Unsuccessful* candidates for the RDD: Eggers and Hainmueller (2009)
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Speeches: Hansard (Proksch and Slapin, 2012; Goet, 2019)

- ▶ 3,144,368 raw speech records, 1811–2015
- ▶ Post-1832 and matched to the roster: 2,203,887 speeches, 7,149 MPs
- ▶ Unit of analysis: the **speaker-year**, kept only inside the MP's documented service
- ▶ All oral question time (PMQs, ministerial answers) removed

Union sponsorship (Fourinaies, 2024)

- ▶ Labour Party records, 1918–1996; about 1,000 sponsored candidates over 20 election cycles

Matching

Summary stats

Who counts as working class?

- ▶ **Source:** History of Parliament roster: *formative* and *pre-election* occupation, each with a Goldthorpe class code (I-VII); education level; typed union connections

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- ▶ **Backstop** for the $\approx 4\%$ uncoded: pre-election code $\rightarrow$ formative code $\rightarrow$ occupation text $\rightarrow$ legislatoR profession; union rule at every tier

Measuring parliamentary attention

- ▶ **CAP keywords** — keyword rules on the CAP taxonomy, applied to debate headings (external anchor)
- ▶ **Structural Topic Model** ($K=28$, PMQ-free) (Roberts et al., 2013) — continuous θ_{dk} per MP-year;
prevalence $\sim s(\text{year}) + \text{party}$, *content* $\sim \text{period}$ (6 eras)
- ▶ **K selection** — coherence-exclusivity frontier, content-free search K over $K \in \{25, \dots, 30\}$;
final model refit with the content covariate

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Material priorities of working-class voters

Core WC topics: CAP 5 (Labour & Employment), 13 (Social Welfare)

Core STM topics: 11 (labour/unemployment), 21 (trade unions), 18 (mining/coal)

Outcome: share of an MP-year's speech on core WC topics

STM topic wordclouds: core and selected topics

Topic 11: Unemployment & labour



Topic 21: Trade unions



Topic 17: Housing & tenancy



Topic 18: Mining & coal



Topic 25: Macroeconomics & tax



Topic 19: International affairs



Topic 5: Elections & franchise



Topic 3: Education



Topic 4: Health



Measuring ideology and register

- ▶ **Ideology: Wordfish** (Slapin and Proksch, 2008)
 - ▶ Unsupervised Poisson scaling
 - ▶ Anchoring highest-speech-count MPs to the poles
 - ▶ Estimated *within* each of the six periods , then z-normalised so positions compare across time

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- ▶ **Register:**
 - ▶ **Lexical diversity:** type-token ratio on a standardised 200-token draw
 - ▶ **Word length:** mean syllables per token
 - ▶ **Concreteness:** mean Brysbaert rating (Brysbaert et al., 2014)

Ideology

Register table

Empirical Strategy

A ladder of designs

- ▶ Rung 0 — panel FE. $y_{it} = \beta \mathbf{W}\mathbf{C}_i + \theta_p + \delta_t + \mathbf{X}_i' \gamma + \varepsilon_{it}$.

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► Rung 1 — within-constituency switching.

$y_{ict} = \beta \mathbf{WC}_i + \alpha_{ce} + \delta_t + \theta_p + \varepsilon_{ict}$, with α_{ce} a constituency $\times$ boundary-era fixed effect.

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- ▶ Rung 2 — constituency-level event study of the switch (de Chaisemartin and D'Haultfœuille, 2026).
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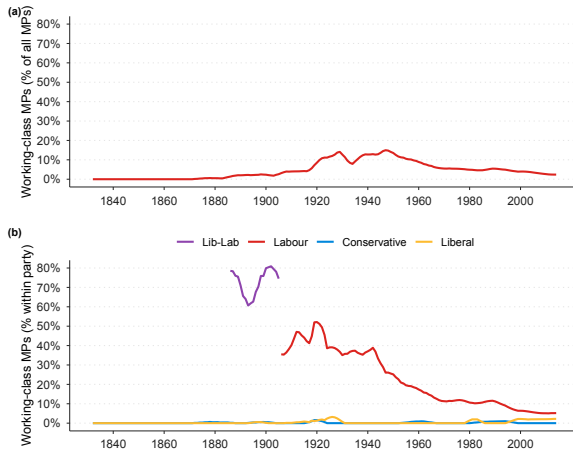
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- ▶ Rung 4 — within-MP union sponsorship. (Fourniaies, 2024) $Y_{it} = \alpha_i + \lambda_t + \tau \text{Sponsor}_{it} + \varepsilon_{it}$

Results

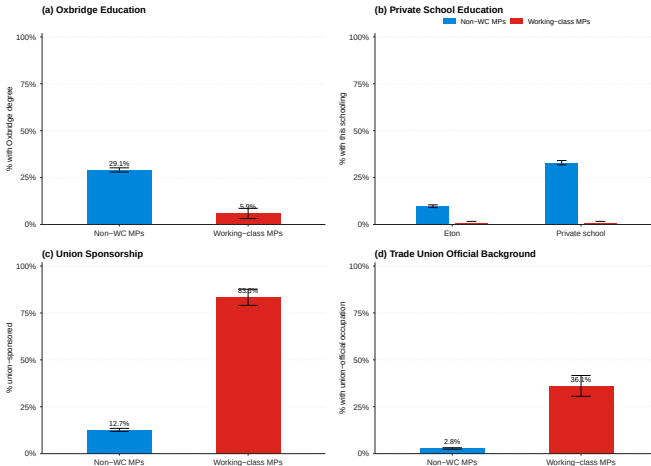
Working-class presence in the House of Commons over time



Notes: Panel (a): share of all MPs classified working-class by the O'Grady (2019) pre-parliamentary rule, by year (5-year moving average). Panel (b): the same share computed separately within Labour, Conservative and Liberal / Liberal Democrat; Lib-Lab MPs shown separately until the Labour series begins (1906). Party-years with fewer than 5 MPs omitted.

Where do working-class MPs come from?

Validation of Working-Class Background Classification
Correlations with Education and Prior Employment



Notes: Prevalence of each validation indicator among working-class (red) and non-working-class (blue) MPs, with 95% confidence intervals. Panel (a) Oxbridge degree; (b) private school and Eton; (c) union sponsorship; (d) prior trade-union official background.

Rung 0: topic allocation and ideology

$$y_{it} = \beta \text{WC}_i + \theta_p + \delta_t + X_i' \gamma + \varepsilon_{it}$$

- ▶ y_{it} : MP i 's core WC topic share (debate headings / topic model) or Wordfish position in year t
- ▶ WC_i : pre-parliamentary working-class background; θ_p party FE, δ_t year FE
- ▶ X_i : female, university, private school, aristocratic connections; + entry-cohort FE
- ▶ SEs clustered by MP

Rung 0: topic allocation and ideology

	(1)
	OLS
<i>Panel A: Debate Headings</i>	
Working-class	0.073*** (0.006)
<i>Panel B: Topic Model</i>	
Working-class	0.113*** (0.008)
<i>Panel C: Wordfish ideology (θ_z)</i>	
Working-class	-0.144*** (0.048)

Notes: Unit of observation is the MP-year, on the common sample of in-service MP-years (51,680 in every column). Party fixed effects: seat party group. Controls: female, university, private school, aristocratic connections. Cohort fixed effects are for the decade of the MP's first election. Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[Full table](#)[Trends by class](#)[Leave-one-out](#)[+ Secondary](#)[Definitions](#)[Backstop](#)[Participation](#)[Table](#)

Rung 0: topic allocation and ideology

	(1) OLS	(2) + Party, Year FE
<i>Panel A: Debate Headings</i>		
Working-class	0.073*** (0.006)	0.049*** (0.006)
<i>Panel B: Topic Model</i>		
Working-class	0.113*** (0.008)	0.085*** (0.008)
<i>Panel C: Wordfish ideology (θ_z)</i>		
Working-class	-0.144*** (0.048)	-0.021 (0.052)

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Working-class	0.073*** (0.006)	0.049*** (0.006)	0.046*** (0.006)
<i>Panel B: Topic Model</i>			
Working-class	0.113*** (0.008)	0.085*** (0.008)	0.078*** (0.008)
<i>Panel C: Wordfish ideology (θ_z)</i>			
Working-class	-0.144*** (0.048)	-0.021 (0.052)	-0.076 (0.053)

Notes: Unit of observation is the MP-year, on the common sample of in-service MP-years (51,680 in every column). Party fixed effects: seat party group. Controls: female, university, private school, aristocratic connections. Cohort fixed effects are for the decade of the MP's first election. Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[Full table](#)
[Trends by class](#)
[Leave-one-out](#)
[+ Secondary](#)
[Definitions](#)
[Backstop](#)
[Participation](#)
[Table](#)

Rung 0: topic allocation and ideology

	(1) OLS	(2) + Party, Year FE	(3) + Controls	(4) + Cohort FE
<i>Panel A: Debate Headings</i>				
Working-class	0.073*** (0.006)	0.049*** (0.006)	0.046*** (0.006)	0.046*** (0.006)
<i>Panel B: Topic Model</i>				
Working-class	0.113*** (0.008)	0.085*** (0.008)	0.078*** (0.008)	0.077*** (0.008)
<i>Panel C: Wordfish ideology (θ_z)</i>				
Working-class	-0.144*** (0.048)	-0.021 (0.052)	-0.076 (0.053)	-0.076 (0.053)

Notes: Unit of observation is the MP-year, on the common sample of in-service MP-years (51,680 in every column). Party fixed effects: seat party group. Controls: female, university, private school, aristocratic connections. Cohort fixed effects are for the decade of the MP's first election. Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Rung 1: within-constituency switching in member type

$$y_{ict} = \beta \mathbf{WC}_i + \alpha_{ce} + \delta_t + \theta_p + \varepsilon_{ict}$$

- ▶ y_{ict} : outcome of MP i sitting for constituency c in year t
- ▶ α_{ce} : constituency $\times$ boundary-era FE
- ▶ β identified only from within-seat changes in the *type* of member
- ▶ δ_t year FE, θ_p party FE; SEs clustered by constituency

Rung 1: within-constituency switching in member type

	(1)	(2)
<i>Panel A: Debate Headings</i>		
Working-class	0.057*** (0.006)	0.049*** (0.006)
<i>Panel B: Topic Model</i>		
Working-class	0.106*** (0.009)	0.085*** (0.009)
<i>Panel C: Wordfish ideology (θ_z)</i>		
Working-class	-0.127*** (0.047)	-0.024 (0.049)
Constituency $\times$ era FE		
Party FE		✓
Year FE	✓	✓
Sample	Full	Full
Observations	51,315	51,315

Notes: Unit of observation is the MP-year. Panel A: CAP core share (5, 13). Panel B: STM $K=28$ core share (topics 11, 21, 18). Panel C: within-period z-normalised Wordfish position. Constituency $\times$ era fixed effects interact the seat with the boundary regime between major redistributions (1832–67, 1868–84, 1885–1917, 1918–49, 1950–73, 1974–82, 1983–96, 1997–2009, 2010–). Column 5 restricts to Labour-held seats. Common in-service MP-year sample; constituency and party from the seat record. Standard errors clustered by constituency.

Rung 1: within-constituency switching in member type

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Debate Headings</i>					
Working-class	0.057*** (0.006)	0.049*** (0.006)	0.041*** (0.008)	0.041*** (0.008)	0.044*** (0.011)
<i>Panel B: Topic Model</i>					
Working-class	0.106*** (0.009)	0.085*** (0.009)	0.060*** (0.009)	0.052*** (0.009)	0.054*** (0.011)
<i>Panel C: Wordfish ideology (θ_z)</i>					
Working-class	-0.127*** (0.047)	-0.024 (0.049)	-0.072 (0.062)	-0.029 (0.066)	-0.138 (0.087)
Constituency $\times$ era FE			✓	✓	✓
Party FE		✓		✓	-
Year FE	✓	✓	✓	✓	✓
Sample	Full	Full	Full	Full	Labour
Observations	51,315	51,315	51,315	51,315	16,565

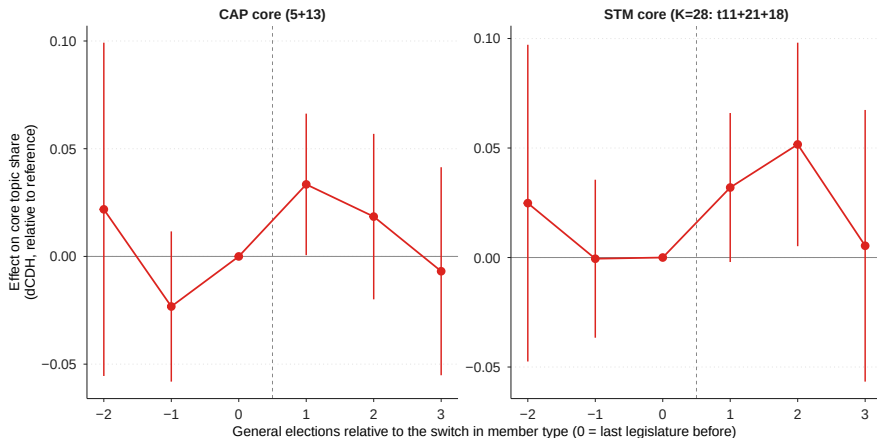
Notes: Unit of observation is the MP-year. Panel A: CAP core share (5, 13). Panel B: STM $K=28$ core share (topics 11, 21, 18). Panel C: within-period z-normalised Wordfish position. Constituency $\times$ era fixed effects interact the seat with the boundary regime between major redistributions (1832–67, 1868–84, 1885–1917, 1918–49, 1950–73, 1974–82, 1983–96, 1997–2009, 2010–). Column 5 restricts to Labour-held seats. Common in-service MP-year sample; constituency and party from the seat record. Standard errors clustered by constituency.

Rung 2: constituency-level event study (de Chaisemartin–D'Haultfoeuille)

$$y_{ce,t} = \sum_{k \neq 0} \beta_k s_{ce} \mathbb{1}[t - (F_{ce} - 1) = k] + \alpha_{ce} + \delta_t + \varepsilon_{ce,t}$$

- ▶ Unit: constituency $\times$ boundary-era cell ce , on the general-election timeline; $y_{ce,t}$ = word-weighted mean core WC share of the legislature
- ▶ F_{ce} = first election at which the seat's WC status switches; $k = 0$ (last legislature before) is the reference
- ▶ $s_{ce} = +1$ for a switch *to*, -1 *away from* a WC member: β_k = effect of the seat becoming WC-held
- ▶ α_{ce} cell FE, δ_t election FE. Estimated heterogeneity-robust

Rung 2: constituency-level event study (de Chaisemartin–D'Haultfœuille)



Notes: de Chaisemartin and D'Haultfœuille (2026) dynamic estimator (`did_multiplert_dyn`); treatment = 1 when the seat is WC-held, pooling switches in both directions. $k = 0$ is the last legislature before the switch. Left: debate headings right: STM $K=28$ core. SEs clustered by constituency $\times$ era.

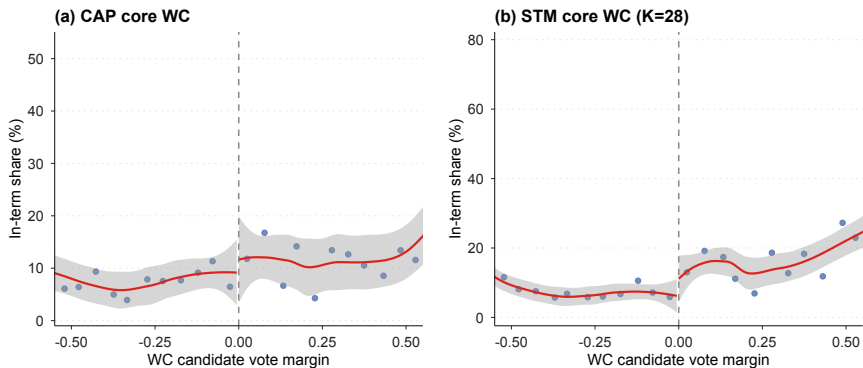
Switching Two-way FE ES Plain-mean placebo

Rung 3: close-election regression discontinuity

$$Y_{ict} = \alpha + \tau \text{Win}_{ict} + f(\text{Margin}_{ict}) + \epsilon_{ict}$$

- ▶ Margin_{ict} : WC candidate's vote margin over the best non-WC opponent; $\text{Win}_{ict} = \mathbb{1}[\text{Margin}_{ict} > 0]$
- ▶ Y_{ict} : winning MP's in-term core WC topic share
- ▶ τ : LATE of *electing* a WC MP at the threshold (Marshall, 2024)
- ▶ $f(\cdot)$ local polynomial, triangular kernel, MSE-optimal bandwidth (rdrobust)

Rung 3: close-election regression discontinuity



Notes: RD of the winning MP's in-term core WC topic share on the WC candidate's margin over the best non-WC opponent (Eggers and Hainmueller (2009) GEs 1950–1970; O'Grady (2019) screen).

Binned means; local-polynomial fits, 95% CIs (rdrobust, triangular kernel, MSE-optimal BW). Left: CAP core; right: STM K=28 core.

[Full table](#) [Kernels](#) [Weighting](#) [Diagnostics](#)

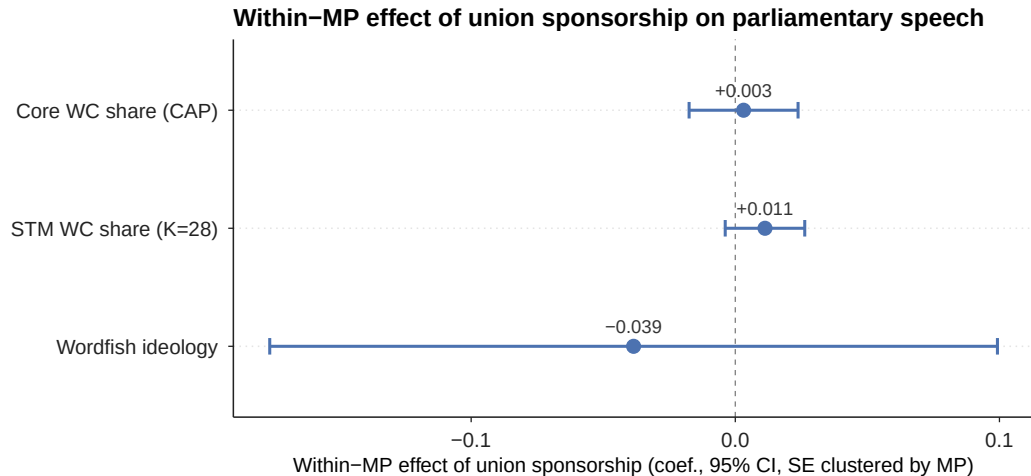
Mechanisms

Rung 4: within-MP effect of union sponsorship

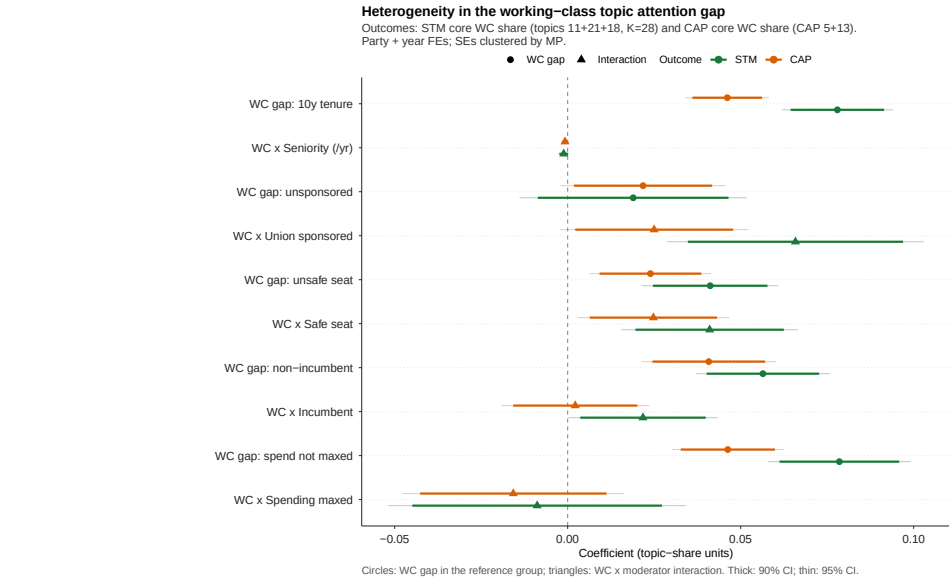
$$Y_{it} = \alpha_i + \lambda_t + \tau \text{Sponsor}_{it} + \varepsilon_{it}$$

- ▶ Sponsor_{it} : Labour MP i is union-sponsored in the term starting at election t (Fourinaies, 2024)
- ▶ α_i MP FE, λ_t Legislature FE

Rung 4: within-MP effect of union sponsorship

[Full table](#)[By WC status](#)[Event study](#)

Heterogeneity by MP and seat characteristics



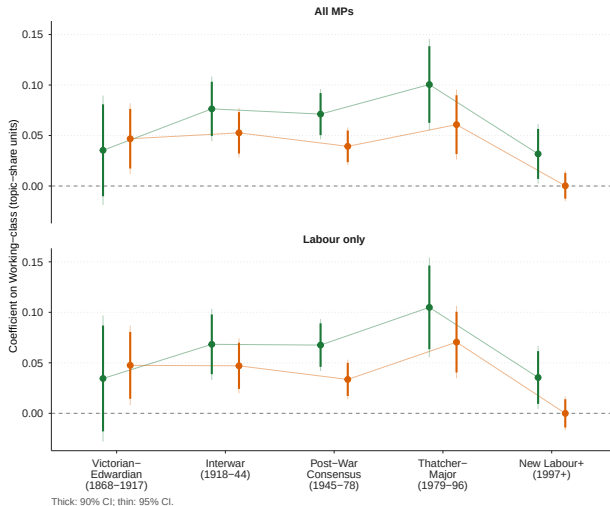
Heterogeneity by historical period

Working-class topic attention gap by historical period

Outcomes: STM core WC share (topics 11+21+18, K=28) and CAP core WC share (CAP 5+13).

All-sample: party + year FEs. Labour-only: year FE. SEs clustered by MP.

Outcome — STM — CAP



How do they speak?

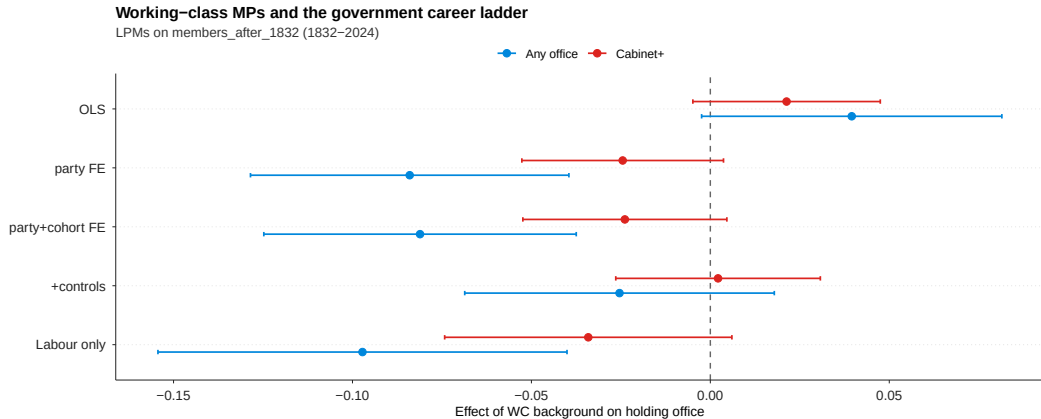
	TTR (lexical diversity)				Syllables per word				Concreteness			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Working-class	-0.030 (0.035)	-0.104*** (0.027)	-0.095*** (0.027)	-0.096*** (0.027)	-0.348*** (0.065)	-0.297*** (0.066)	-0.253*** (0.066)	-0.253*** (0.064)	0.532*** (0.051)	0.275*** (0.047)	0.173*** (0.048)	0.171*** (0.048)
Standard-Errors						Clustered by MP						
Observations	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680
R ²	0.00004	0.20539	0.20578	0.20907	0.00564	0.24272	0.24471	0.25829	0.01318	0.38564	0.40292	0.40795
Raw mean of outcome		0.632				1.525				2.329		
Raw SD of outcome		0.040				0.053				0.050		
Party fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Year fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Individual controls			✓	✓			✓	✓			✓	✓
Decade cohort FEs				✓				✓				✓

Notes: Common in-service MP-year sample (51,680 in every column), Hansard 1832–2014. Each outcome standardised to unit variance, so coefficients on Working-class read in SDs. *Individual controls*: female, university, private (fee-paying) school, aristocratic connections. SEs clustered by MP.

Ideology

Local vs national

The government career ladder



Table

Conclusion

Franchise is necessary but not sufficient

- ▶ Substantive representation of a social group requires that its members *actually sit* in the legislature

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- ▶ Set scope conditions for when to expect elite persistence to matter

Future work: *whom* do they speak for?

→ The unionized or the broad working-class?

Thank you!

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Appendix

A. Data construction

- ▶ Matching cascade
- ▶ Summary statistics
- ▶ Speakers and speeches
- ▶ Party composition
- ▶ Educational background
- ▶ Formative occupations

B. WC classification

- ▶ Background validation grid
- ▶ Education crosstab
- ▶ Definition robustness
- ▶ Leave-one-out topics
- ▶ Backstop robustness
- ▶ Core + secondary topics

C. Text models

- ▶ K-selection frontier
- ▶ Topic proportions
- ▶ Topics over time
- ▶ STM → CAP assignment
- ▶ External validation heatmap
- ▶ Wordclouds; representative speeches
- ▶ Party gap in emphasis
- ▶ CAP salience tiers

D. Additional results

- ▶ Broad topics; core vs broad gap
- ▶ Topic decomposition
- ▶ Speech participation
- ▶ Ideology (four views)
- ▶ Local vs national

E. Designs

- ▶ Constituency switching
- ▶ Switching ES (TWFE, dCdH)
- ▶ RDD full table; kernels; weighting; diagnostics
- ▶ Union sponsorship descriptives
- ▶ Sponsorship full table; by WC
- ▶ Sponsorship ES: cells, dCdH, TWFE

F. Regression tables

- ▶ Topic and ideology
- ▶ Heterogeneity; periods
- ▶ Language complexity
- ▶ Career outcomes and loyalty

A1 — Data construction: matching cascade

Eight-stage cascade (each stage applied only to records not yet matched; exact identifiers first, fuzzy last; seeded at 42)

1. Wikidata ID exact match
2. MNIS ID exact match
3. TheyWorkForYou ID exact match
4. Exact surname, first name and birth year
5. Fuzzy surname (≤ 1), exact birth year, first initial
6. Fuzzy surname (≤ 1), fuzzy first name (≤ 2), birth year ± 2
7. legislatoR to Eggers-Spirling member ID
8. Hansard speaker field to roster, fuzzy name similarity

Sample: speaker-years kept only inside the matched MP's documented seat record (± 1 year; Fourniaies wins for 341 members without one): **51,680 MP-years** in the common sample

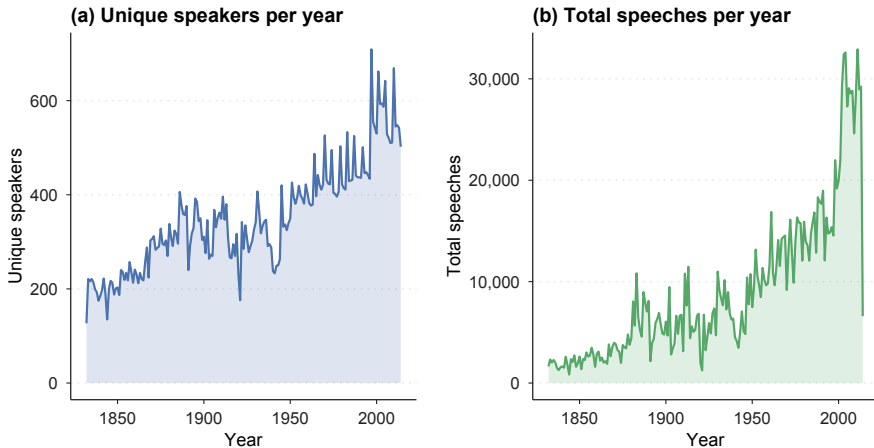
Constituency $\times$ election panel

- ▶ Tenure expanded to one row per MP, constituency and general election (winners only): **26,666 records, 9,990 MPs**
- ▶ Constituency names harmonised (case-folded, token-sorted, directional and county-suffix variants) and matched on name and year; fuzzy matching *deliberately avoided*
- ▶ Candidate names recovered by aligning vote totals and party within constituency-year
- ▶ In the 1885–2017 GB window, 88% of records carry the Cagé and Dewitte (2025) electoral and census variables; 70% carry the MP's own margin

A2 — Summary statistics: speeches dataset

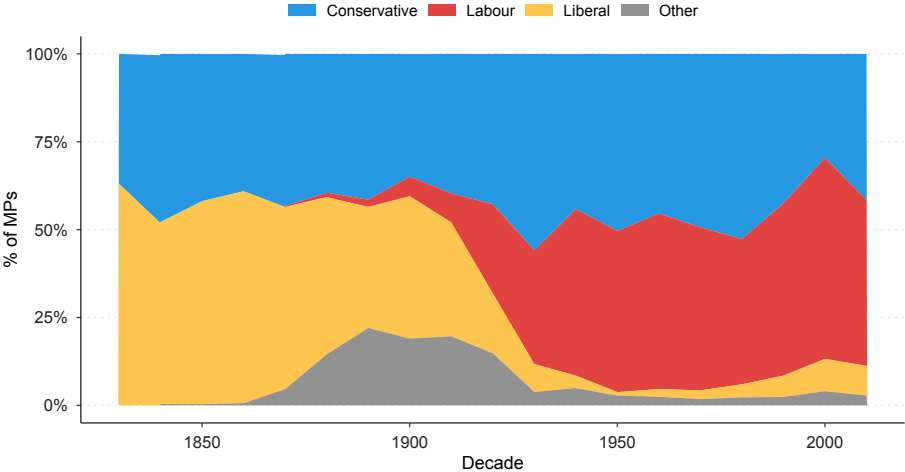
Statistic	Value
Speaker-year observations	63,788
Unique MPs	6,624
Time span	1832–2014
Working-class MPs	288 (4.3%)
Union-sponsored MPs	1,042 (15.7%)
Mean speeches per speaker-year	25.7
<i>Party breakdown (unique MPs)</i>	
Conservative	2,888 (43.6%)
Liberal	1,803 (27.2%)
Labour	1,471 (22.2%)
Other	462 (7.0%)

A3 — Unique speakers and total speeches over time

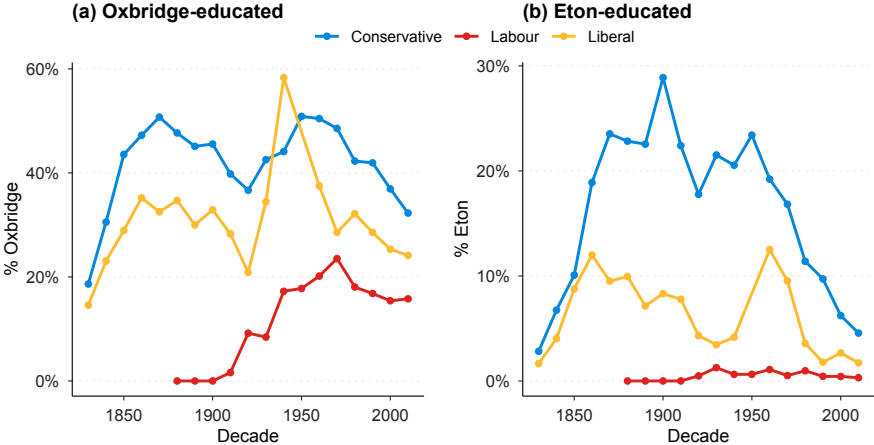


Notes: Panel (a): number of unique MPs making at least one speech per year. Panel (b): total speeches per year. In-service MP-years, Hansard 1832–2014.

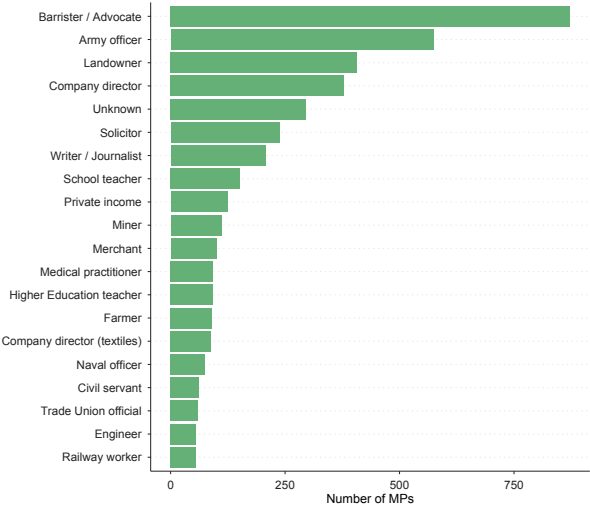
A4 — Party composition of the House of Commons



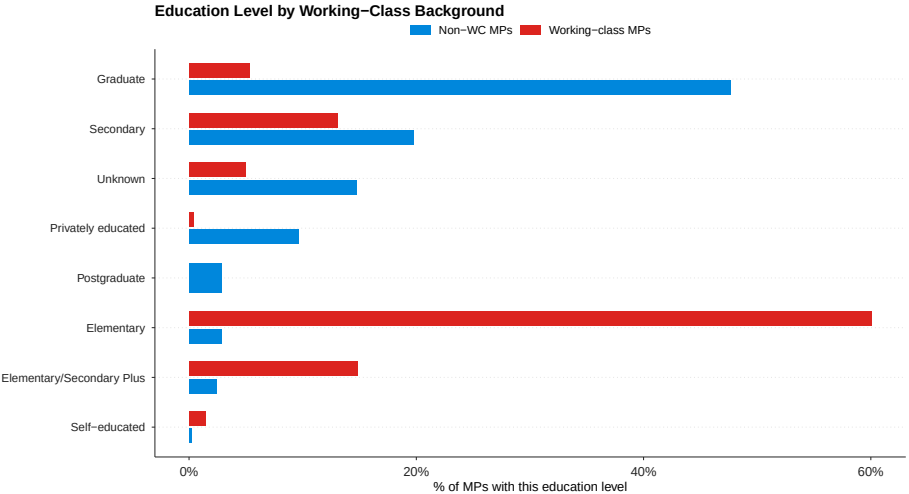
A5 — Educational background of MPs by party



A6 — Formative occupation of MPs (top 20)



B1 — Education level by working-class background



Notes: Distribution of education levels among working-class (red) and non-working-class (blue) MPs. Categories are ordered by prevalence among non-working-class MPs. The working-class indicator is built on occupation, not education.

B2 — Working-class definition robustness

Table 1: Working-Class Definition Robustness: STM Topic-Attention Gap (K=28)

	(1) OLS	(2) +Party, Year FE	(3) +Controls	(4) +Cohort	Obs.
Text hand-list (formative)	0.089*** (0.006)	0.063*** (0.006)	0.058*** (0.006)	0.060*** (0.006)	51,680
Text hand-list (either)	0.087*** (0.005)	0.063*** (0.005)	0.057*** (0.005)	0.058*** (0.005)	51,680
Goldthorpe V-VII {5,6,7}	0.104*** (0.006)	0.081*** (0.006)	0.076*** (0.006)	0.076*** (0.006)	51,680
Goldthorpe manual {6,7}	0.111*** (0.007)	0.087*** (0.007)	0.081*** (0.007)	0.081*** (0.007)	51,680
Goldthorpe broad {3,5,6,7}	0.085*** (0.005)	0.062*** (0.005)	0.056*** (0.005)	0.057*** (0.005)	51,680
Hybrid (code→text→leg.)	0.105*** (0.006)	0.082*** (0.006)	0.077*** (0.006)	0.077*** (0.006)	51,680
O'Grady {3ng,6,7}+union	0.116*** (0.008)	0.085*** (0.008)	0.077*** (0.008)	0.077*** (0.008)	45,866
legislatoR profession	0.086*** (0.008)	0.053*** (0.008)	0.046*** (0.008)	0.046*** (0.008)	51,680
Party fixed effects		✓	✓	✓	
Year fixed effects		✓	✓	✓	
Controls			✓	✓	
Decade cohort FEs				✓	

Outcome: STM K=28 core WC topic share (topics 11+21+18). Each row uses a different MP-level working-class indicator from the definition comparison (03d): text hand-list, Goldthorpe-class codes at three boundaries, the coalesced hybrid, the O'Grady pre-parliamentary rule, and the legislatoR profession layer. Table ?? uses the O'Grady rule with a coverage backstop for MPs without a pre-election Goldthorpe code; the O'Grady row here is the pure rule without the backstop.

Decade cohort FEs (column 4) are fixed effects for the decade of the MP's first election, absorbing differences across generations of entrants.

Obs. is the estimation sample, identical across columns 1–4. All rows use the common MP-year sample of Table ?? (columns 1–12); in the WC-definition and backstop tables it is further intersected with each indicator's coverage (fixed-effect singletons removed). Party fixed effects are for the party group under which the MP held their seat in that year.

Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B3 — Leave-one-out over core STM topics

Table 2: Leave-One-Out over Core STM Topics: Working-Class Topic-Attention Gap (K=28)

	(1) OLS	(2) +Party, Year FE	(3) +Controls	(4) +Cohort	Obs.
Full core (11+21+18)	0.113*** (0.008)	0.085*** (0.008)	0.078*** (0.008)	0.077*** (0.008)	51,680
drop t11 (labour/unemployment)	0.081*** (0.007)	0.063*** (0.007)	0.059*** (0.007)	0.059*** (0.007)	51,680
drop t21 (trade unions)	0.081*** (0.008)	0.063*** (0.007)	0.058*** (0.008)	0.058*** (0.008)	51,680
drop t18 (mining/coal)	0.064*** (0.005)	0.044*** (0.005)	0.038*** (0.005)	0.038*** (0.005)	51,680
Party fixed effects		✓	✓	✓	
Year fixed effects		✓	✓	✓	
Controls			✓	✓	
Decade cohort FEs				✓	

Outcome: STM K=28 core WC topic share. Each row drops one of the three core topics; the coefficient is on Working-class (the O'Grady (2019) pre-parliamentary rule with a coverage backstop, as in Table ??).

Decade cohort FEs (column 4) are fixed effects for the decade of the MP's first election, absorbing differences across generations of entrants.

Obs. is the estimation sample, identical across columns 1–4. All rows use the common MP-year sample of Table ?? (columns 1–12); in the WC-definition and backstop tables it is further intersected with each indicator's coverage (fixed-effect singletons removed). Party fixed effects are for the party group under which the MP held their seat in that year.

Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B4 — Dropping backstop-imputed MPs

Table 3: Robustness: Working-Class Topic-Attention Gap Dropping Backstop-Imputed MPs (K=28)

	(1) OLS	(2) +Party, Year FE	(3) +Controls	(4) +Cohort	Obs.
Full measure (O’Grady + backstop)	0.113*** (0.008)	0.085*** (0.008)	0.078*** (0.008)	0.077*** (0.008)	51,680
Pure O’Grady (backstop dropped)	0.116*** (0.008)	0.085*** (0.008)	0.077*** (0.008)	0.077*** (0.008)	45,866
Party fixed effects		✓	✓	✓	
Year fixed effects		✓	✓	✓	
Controls			✓	✓	
Decade cohort FEs				✓	

Outcome: STM K=28 core WC topic share (topics 11+21+18). Working-class is the paper-wide O’Grady pre-election screen with a coverage backstop. Row 1 uses the full measure; Row 2 drops MPs classified only by the backstop (no pre-election Goldthorpe code, so the pure `wc_ograde` rule is missing) – the cases whose candidate-stage classification cannot be independently verified.

Decade cohort FEs (column 4) are fixed effects for the decade of the MP’s first election, absorbing differences across generations of entrants.

Obs. is the estimation sample, identical across columns 1–4. All rows use the common MP-year sample of Table ?? (columns 1–12); in the WC-definition and backstop tables it is further intersected with each indicator’s coverage (fixed-effect singletons removed). Party fixed effects are for the party group under which the MP held their seat in that year.

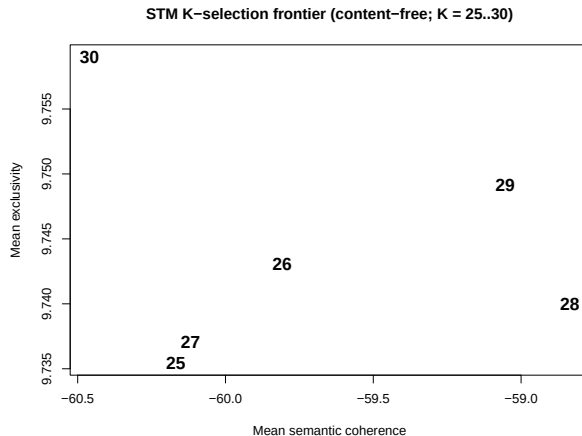
Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B5 — Core + strong-secondary working-class topics

	(1) OLS	(2) + Party, Year FE	(3) + Controls	(4) + Cohort FE
<i>Panel A: Debate headings (CAP 5, 13 + 14, 3, 6, 1)</i>				
Working-class	0.061*** (0.007)	0.033*** (0.008)	0.035*** (0.008)	0.035*** (0.008)
<i>Panel B: Topic model (STM 11, 21, 18 + 17, 4, 25)</i>				
Working-class	0.124*** (0.009)	0.086*** (0.009)	0.080*** (0.010)	0.080*** (0.010)
Observations	51,680	51,680	51,680	51,680

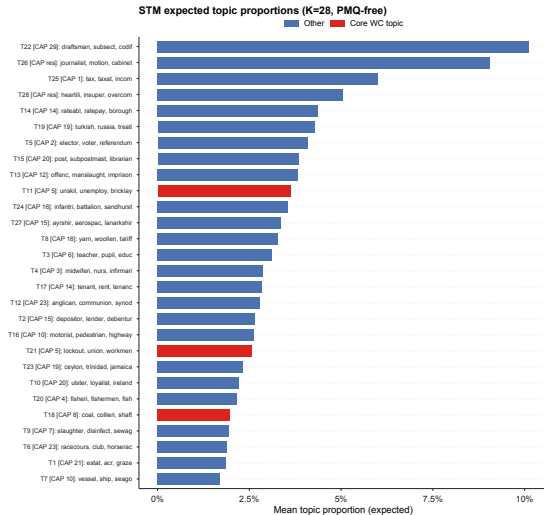
Notes: Outcome is the share of an MP-year's speech on core plus strong-secondary working-class topics. Panel A: CAP-keyword debate-heading shares, core (5 Labour & Employment, 13 Social Welfare) plus strong secondary (14 Housing, 3 Health, 6 Education, 1 Macroeconomics). Panel B: STM ($K=28$, PMQ-free) probabilities, core (11 labour/unemployment, 21 trade unions, 18 mining/coal) plus 17 (housing/tenancy), 4 (health/hospitals) and 25 (taxation/budget); no STM topic is assigned to Education. Same common sample of in-service MP-years and specifications as the Rung 0 table: party FE = seat party group; controls: female, university, private school, aristocratic connections; cohort FE = decade of first election. Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C1 — STM K -selection frontier (content-free)

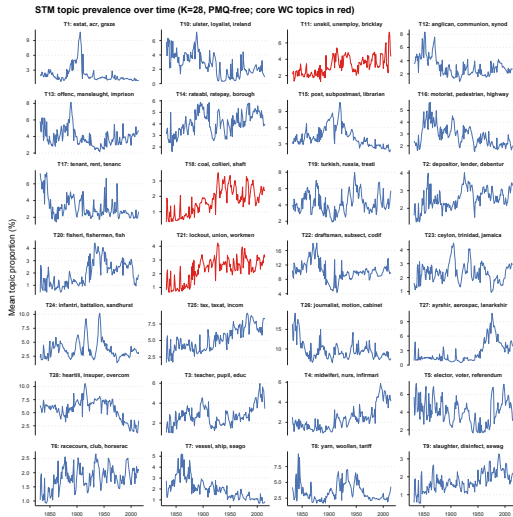


Notes: Semantic coherence against exclusivity for $K \in \{25, \dots, 30\}$ from a content-free `searchK`. Exclusivity is undefined under a content-covariate model, so K is selected here and the final model is then refit with the content covariate.

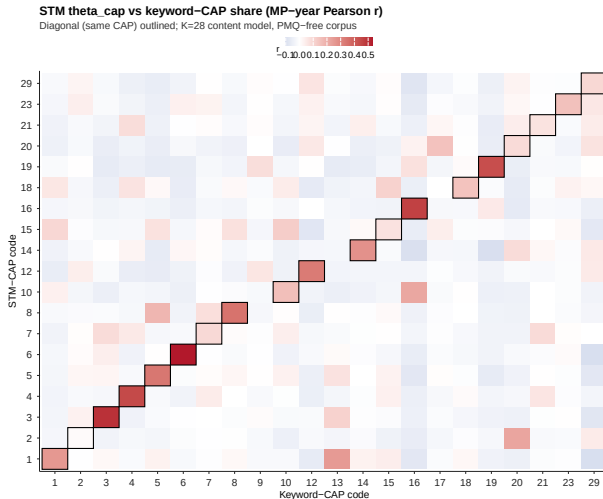
C2 — STM topic proportions (K=28)



C3 — STM topic prevalence over time (K=28)



C4 — External validation of the topic measure



Notes: Panel A correlates, at the MP-year level, each STM CAP-aggregated topic share with the independent CAP-keyword share for the same category.

C5 — STM topic wordclouds: all 28 topics ($K=28$)



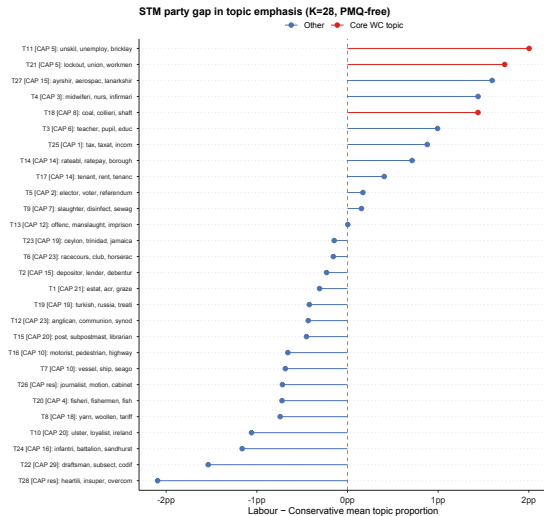
Notes: Highest-probability stemmed words for all 28 topics of the final PMQ-free STM. Word size is proportional to within-topic probability.

C6 — Representative speeches by STM topic

K=28 Topic 1: estat, acr, graze, landown, rentcharg, valuer, reclam,

asked whether the estates commissioners have received a communication from over forty tenants of uneconomic holdings on the de clifford estate, county mayo, objecting to a sale of any portion of the property except through the congested districts board whether, in the event of a sale to the estates commissioners, the commissioners will acquire the nonresidential and other large farms necessary for enlargement of the uneconomic holdings whether one tenant, paying 300 a year rent on this estate, owns a large farm on another property which he acquired under the land purchase acts and what steps the commissioners and the board intend taking to relieve the congestion that exists over a large portion of this estate asked when mr fitzgeraldkenny lodged with the congested districts board maps and documents relating to his congested estate near ballyglass, county mayo, with a view to a sale thereof through the board have the board made an offer to mr fitzgeraldkenny for the estate and, if so, has it been accepted, and what has caused the delay in the negotiations and the completion of purchase asked whether the estates commissioners received within the prescribed time an application from pat heneghan now deceased for reinstatement in a holding at millerhill, on the blossie estate, county mayo, from which he had been evicted and will the commissioners take steps to reinstate his daughter and legal representative, julia heneghan, or to provide her with an equivalent holding asked whe

C7 — STM party gap in topic emphasis (Labour — Conservative)

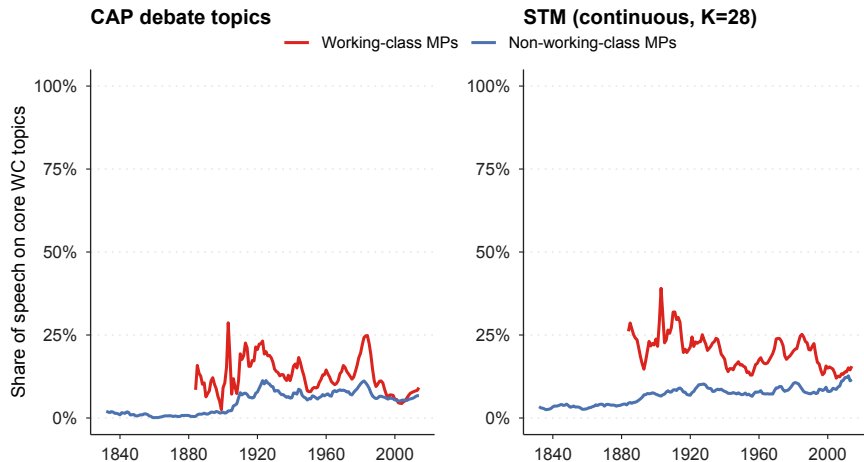


C8 — Working-class salience tiers of CAP major topics

Category	Included CAP topics	Notes
Core working-class	5, 13	Labour, Welfare
Strong secondary	14, 3, 6, 1	Housing, Health, Education, Macroeconomy
Contextual / tertiary	8, 10, 15, 18	Cost-of-living, trade-exposed, or infrastructure
Excluded	2, 4, 7, 9, 12, 16–23, 29	Weak or neutral class salience

Core topics are those most consistently identified as material priorities of working-class voters in survey evidence (Evans and Tilley, 2017; O'Grady, 2019; Heath, 2015).

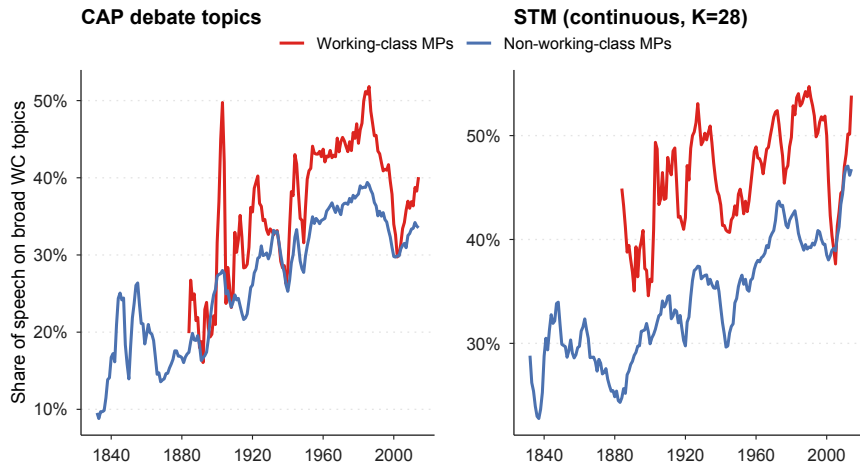
Do — Core working-class topic attention by background



Notes: Share of parliamentary speech devoted to core working-class topics by working-class and non-working-class MPs, 5-year moving average. Left panel: CAP-keyword debate-topic shares (CAP 5, 13). Right panel: STM continuous topic probabilities ($K=28$, topics 11, 21, 18).

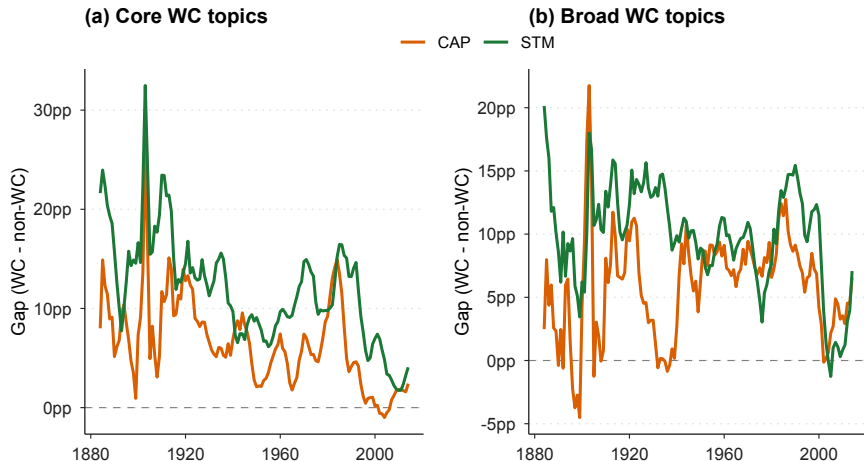
Broad topics Core vs broad gap Decomposition

D1 — Broad working-class topic attention by background



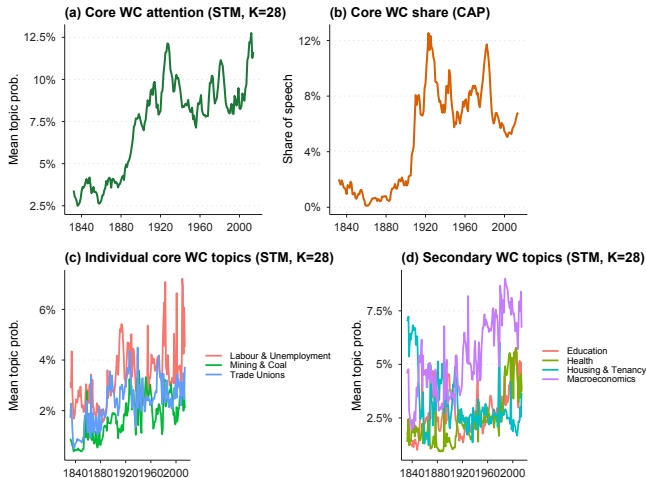
Notes: Share of speech on the broad set of working-class topics (core plus Housing, Health, Education, Macroeconomics, Energy, Transportation, Banking & Commerce, Foreign Trade), 5-year moving average. Left: CAP-keyword shares. Right: STM K=28 probabilities.

D2 — Robustness: the attention gap from core to broad topics



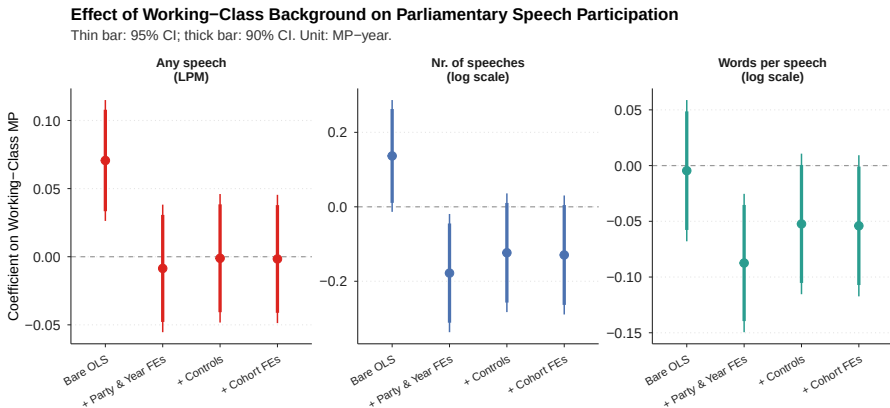
Notes: Difference in topic attention between working-class and non-working-class MPs (WC minus non-WC), 5-year moving average. Panel (a): core topics (CAP 5, 13; STM $K=28$ topics 11, 21, 18). Panel (b): broad topics (core plus Housing, Health, Education, Macroeconomics, Energy, Transportation, Banking & Commerce, Foreign Trade). Shown for the CAP-keyword and STM ($K=28$) measures.

D3 — Working-class topic attention decomposed



Notes: Panel (a): aggregate core WC topic probability from the STM ($K=28$). Panel (b): CAP-keyword hard classification share for core WC topics. Panels (c) and (d): individual core and secondary WC topics from the STM, 5-year moving average.

D4 — Working-class background and speech participation



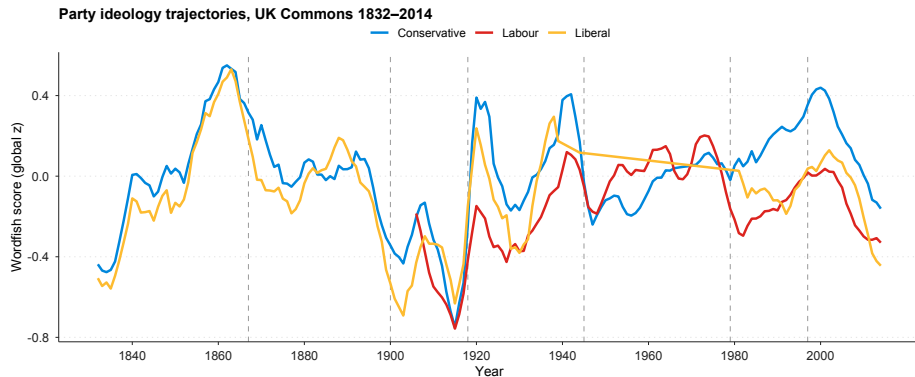
Notes: OLS coefficients on Working-class for three outcomes in the balanced MP-year panel (every MP-year covered by a seat record, 1832–2014, $N = 125,994$; outcome 3 conditions on at least one speech, $N = 63,088$). Specifications: (1) bare OLS; (2) seat-party and year fixed effects; (3) adds the individual controls; (4) adds decade-of-first-election cohort fixed effects. Thin bars 95% CI, thick bars 90% CI.

D5 — Speech participation: full regression table

	Any Speech (LPM)				Log Nr. Speeches				Log Words/Speech			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Working-class	0.071*** (0.023)	-0.009 (0.024)	-0.001 (0.024)	-0.002 (0.024)	0.137* (0.077)	-0.178** (0.081)	-0.123 (0.081)	-0.129 (0.082)	-0.005 (0.032)	-0.087*** (0.032)	-0.052 (0.032)	-0.054* (0.032)
Female			-0.086*** (0.021)	-0.088*** (0.022)			-0.306*** (0.073)	-0.295*** (0.074)			0.012 (0.022)	-0.004 (0.022)
University			0.040*** (0.009)	0.038*** (0.009)			0.196*** (0.029)	0.189*** (0.029)			0.078*** (0.013)	0.080*** (0.013)
Private school			-0.016 (0.016)	-0.016 (0.016)			-0.051 (0.040)	-0.057 (0.040)			-0.089** (0.036)	-0.086** (0.036)
Aristocratic			-0.074*** (0.012)	-0.075*** (0.012)			-0.157*** (0.035)	-0.173*** (0.034)			0.051* (0.022)	0.048** (0.022)
Standard-Errors	Clustered by MP											
Observations	125,994	125,994	125,994	125,994	125,994	125,994	125,994	125,994	63,088	63,088	63,088	63,088
R ²	0.0008	0.1079	0.1130	0.1161	0.0003	0.2008	0.2071	0.2116	0.0000	0.0876	0.0920	0.0960
Party fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Year fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Decade cohort FEs				✓				✓				✓

Notes: Columns 1–4 Any Speech (linear probability model); 5–8 log(1 + n_{speeches}); 9–12 log words per speech. Standard errors clustered by MP.

D6 — Party ideology and working-class representation



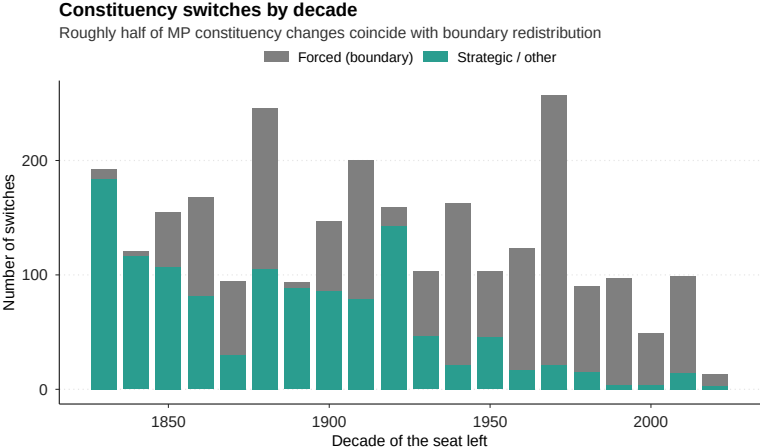
Notes: Mean Wordfish score by party (z-normalised within period; higher is more right-leaning), 5-year moving average. Dashed vertical lines mark major institutional milestones.

D7 — Constituency vs national speech orientation

	Localism Share (SD)				Own-Constituency Name (SD)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Working-class	-0.036 (0.051)	0.076* (0.040)	0.043 (0.041)	0.040 (0.040)	0.054 (0.049)	0.100* (0.052)	0.067 (0.054)	0.068 (0.055)
Female			0.230*** (0.039)	0.139*** (0.036)			0.138*** (0.069)	0.042 (0.069)
University			-0.098*** (0.017)	-0.091*** (0.017)			-0.086*** (0.022)	-0.079*** (0.022)
Private school			0.005 (0.039)	0.017 (0.038)			0.029 (0.035)	0.038 (0.035)
Aristocratic			-0.106*** (0.023)	-0.073*** (0.023)			-0.107*** (0.016)	-0.087*** (0.016)
Standard-Errors	Clustered by MP							
Observations	49,774	49,774	49,774	49,774	49,774	49,774	49,774	49,774
R ²	0.00006	0.19093	0.19664	0.21570	0.00014	0.02701	0.03084	0.04790
Party fixed effects		✓	✓	✓		✓	✓	✓
Year fixed effects		✓	✓	✓		✓	✓	✓
Decade cohort FEs				✓				✓

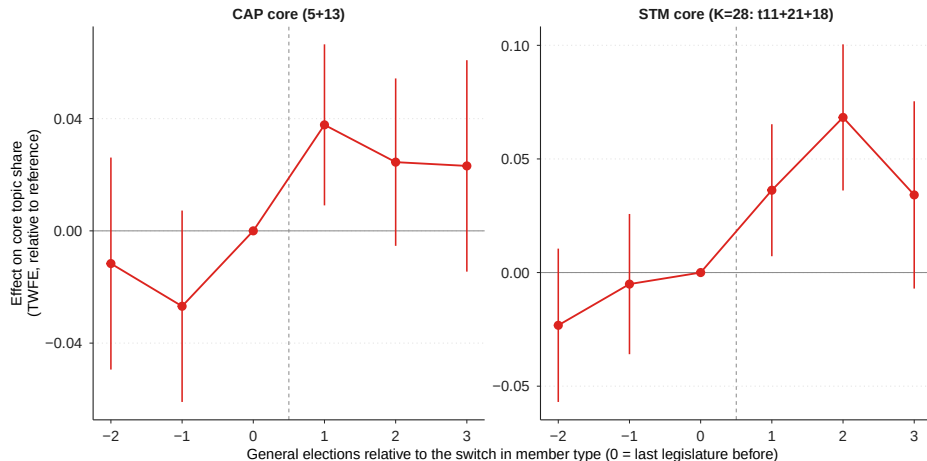
Notes: *Localism share* = local / (local + national) dictionary hits, where local terms are constituency-reference words plus the MP's own year-specific seat name(s) and national terms are nation/country/kingdom/empire words. *Own-constituency name* is own-seat mentions per 1,000 spoken words. Both outcomes are standardised. Standard errors clustered by MP.

E1 — Constituency switching by MPs over time



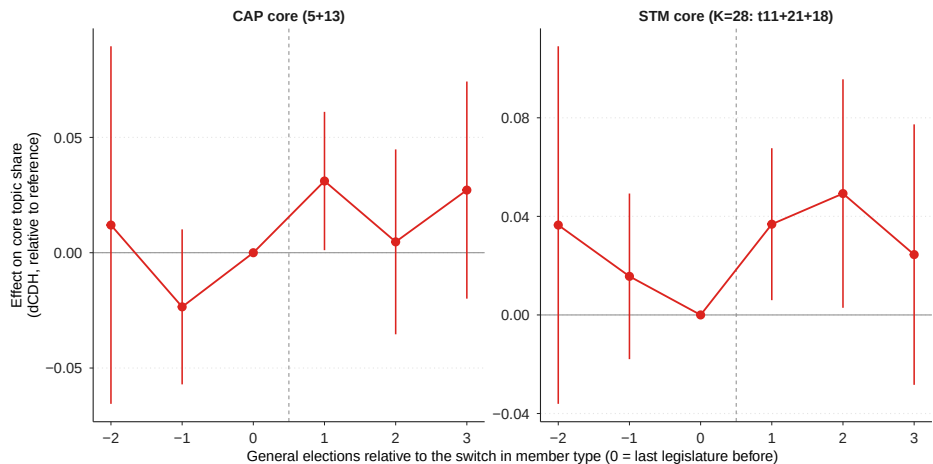
Notes: Number of MP constituency switches per decade, by the decade of the seat departed (History of Parliament seats data, 1832–2024; 2,673 switches among 2,193 MPs). Switches are classified as boundary-forced — the prior constituency was abolished, or the move occurred at a redistribution election — versus strategic or other.

E1b — Switching event study: two-way fixed effects



Notes: Pooled two-way fixed-effects event study on the same constituency $\times$ era $\times$ general-election panel, outcome and event-time axis as the dCdH figure: $k = 0$ is the last legislature before the switch (reference), $k = -2, -1$ pre-switch, $k = 1, 2, 3$ post-switch. Switches away from a working-class member enter with the opposite sign, so coefficients are the effect of the seat becoming working-class held. Constituency $\times$ era and general-election fixed effects; never-switching cells as controls; SEs clustered by cell. Left: CAP core share. Right: STM $K=28$ core share. Exposed to the negative-weighting critique of staggered/reversible designs; the heterogeneity-robust counterpart is the main-text dCdH figure.

E1c — Switching event study: plain-mean placebo (dCdH)



Notes: de Chaisemartin and D'Haultfoeuille (2026) dynamic estimator with the outcome measured as the *plain* legislature mean core-topic share (rather than word-weighted). Average effects +0.025 (CAP, SE 0.015) and +0.043 (STM, SE 0.017); joint-placebo $p = 0.37$ and $p = 0.51$, as with the word-weighted outcome. Left: CAP core; right: STM K=28 core.

E2 — Close-election RDD: full table

Table 4: Close-Election RDD (Eggers–Hainmueller 1950–1970, new K=28 STM)

Outcome	h	Coef.	Coef. (SD)	SE	p (rob.)	95% CI	N_l/N_r
<i>All working-class candidates (any party) (344 elections)</i>							
Debate Headings	0.252	+0.0251	+0.165	0.0596	0.646	[-0.089, 0.144]	101/88
Topic Model	0.118	+0.0272	+0.211	0.0344	0.700	[-0.054, 0.081]	48/46
STM t11: Labour/unemployment	0.114	+0.0092	+0.141	0.0170	0.987	[-0.034, 0.033]	48/45
STM t21: Trade unions	0.175	+0.0295*	+0.444	0.0177	0.093	[-0.005, 0.064]	66/70
STM t18: Mining/coal	0.131	-0.0081	-0.112	0.0144	0.395	[-0.040, 0.016]	49/53
<i>Labour working-class candidates only (255 elections)</i>							
Debate Headings	0.141	+0.0035	+0.023	0.0729	0.852	[-0.157, 0.129]	48/56
Topic Model	0.114	+0.0309	+0.238	0.0368	0.586	[-0.052, 0.092]	40/44
STM t11: Labour/unemployment	0.108	+0.0159	+0.233	0.0184	0.721	[-0.029, 0.043]	39/43
STM t21: Trade unions	0.155	+0.0255	+0.445	0.0197	0.289	[-0.018, 0.060]	49/59
STM t18: Mining/coal	0.111	-0.0085	-0.108	0.0171	0.716	[-0.040, 0.027]	40/43

Triangular kernel, MSE-optimal bandwidth (`rdrobust`, `mserd`); robust bias-corrected p -values and 95% CIs. Sample: Eggers–Hainmueller (2009) UK general elections 1950–1970 with exactly one working-class candidate (O’Grady headline screen) against a non-working-class opponent; running variable is the working-class candidate’s vote margin. **Outcome is the winning MP’s word-weighted average speech share over the parliamentary term following the election** (calendar years from the year after the election to the year before the next general election; MP-years weighted by total words spoken +1, as in the constituency-switching design); elections whose winner has no qualifying in-term speech-year are dropped. Topic Model (STM) core WC = K=28 topics 11+21+18; Debate Headings (CAP) core WC = CAP-keyword shares 5+13. Coef. (SD) divides by the outcome SD. N_l/N_r = effective observations left/right of the cutoff. Kernel, bandwidth, and weighting robustness in Table ??.

E2b — Close-election RDD: kernel and bandwidth

Specification	Debate Headings		Topic Model	
	All WC	Labour WC	All WC	Labour WC
Triangular, MSE-opt.	+0.0251 (0.0596)	+0.0035 (0.0729)	+0.0272 (0.0344)	+0.0309 (0.0368)
Uniform, MSE-opt.	+0.0210 (0.0606)	+0.0161 (0.0709)	+0.0365 (0.0358)	+0.0328 (0.0475)
Epanechnikov, MSE-opt.	+0.0263 (0.0606)	+0.0027 (0.0783)	+0.0276 (0.0366)	+0.0326 (0.0400)
Triangular, CER-opt.	+0.0269 (0.0602)	-0.0002 (0.0707)	+0.0283 (0.0320)	+0.0357 (0.0343)
Triangular, $h = 5pp$	-0.0791 (0.1813)	-0.0817 (0.1935)	+0.0051 (0.0432)	+0.0104 (0.0470)
Triangular, $h = 10pp$	+0.0340 (0.0844)	+0.0057 (0.0864)	+0.0270 (0.0332)	+0.0327 (0.0346)
Triangular, $h = 15pp$	+0.0201 (0.0737)	+0.0075 (0.0715)	+0.0380 (0.0345)	+0.0380 (0.0362)
Triangular, $h = 20pp$	+0.0252 (0.0687)	+0.0299 (0.0709)	+0.0521 (0.0329)	+0.0531 (0.0347)

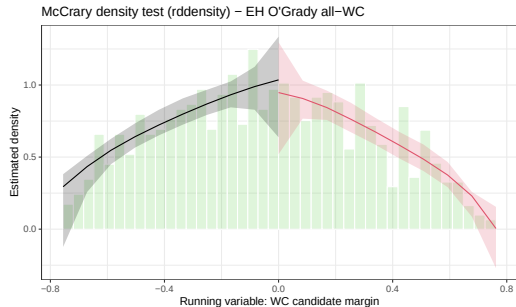
Notes: Coefficient (robust SE) on winning, from `rdzrobust`; no estimate is significant at the 10% level (robust bias-corrected p). Outcome: the winning MP's word-weighted core WC share over the parliamentary term following the election. Rows vary the kernel, MSE- vs CER-optimal bandwidth, and fixed windows of ± 5 – $20pp$ of vote margin. Debate Headings: CAP 5+13; Topic Model: STM $K=28$ topics 11+21+18. 344 elections (all WC), 255 (Labour WC) with an STM outcome.

E2c — Close-election RDD: weighting the parliamentary term

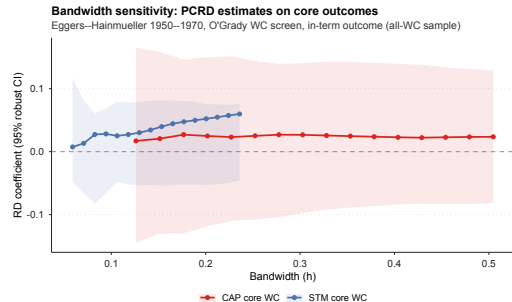
Specification	Debate Headings		Topic Model	
	All WC	Labour WC	All WC	Labour WC
Word-weighted term mean (headline)	+0.0251 (0.0596)	+0.0035 (0.0729)	+0.0272 (0.0344)	+0.0309 (0.0368)
Unweighted term mean	+0.0294 (0.0594)	+0.0133 (0.0717)	+0.0141 (0.0358)	+0.0192 (0.0387)

Notes: Coefficient (robust SE) on winning; triangular kernel, MSE-optimal bandwidth (`rdrobust`). The outcome averages the winning MP's core WC share over the calendar years of the following parliamentary term (year after the election to year before the next general election). *Word-weighted*: MP-years weighted by words spoken +1, as in the switching design (headline). *Unweighted*: plain mean over in-term MP-years. Debate Headings: CAP 5+13; Topic Model: STM K=28 topics 11+21+18.

E3 — Close-election RDD: diagnostics



Notes: McCrary density test of the running variable at the electoral threshold.



Notes: Point estimate and 95% CI for the core working-class topic share across a range of bandwidths around the MSE-optimal choice.

E4 — Descriptives by union sponsorship (Labour MP-years)

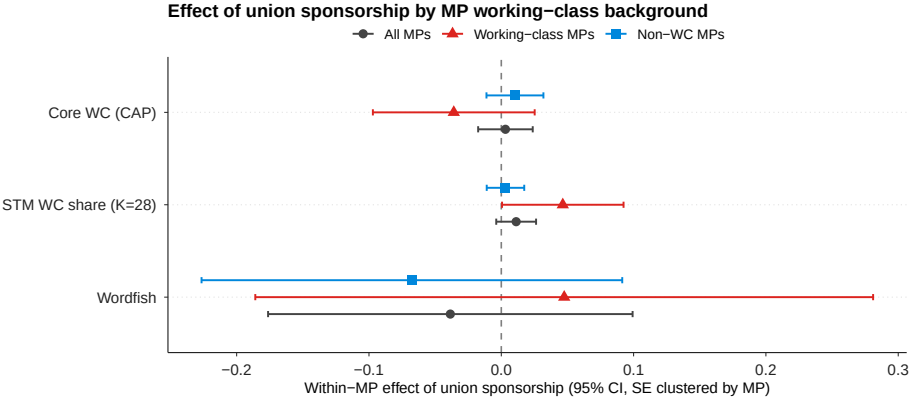
	MP-years	% WC	WC topics	Legal topics	Wordfish
Union-sponsored MPs	11,305	19.2%	9.1%	8.5%	-0.04
Other Labour MPs	5,414	4.4%	6.8%	9.5%	+0.03

E5 — Within-MP effect of sponsorship: full table

Outcome	$\hat{\tau}$	(SE)	N
<i>Electoral outcomes</i>			
Win (0/1)	+0.1863***	(0.0200)	12,755
Vote share (%)	+5.9367***	(0.5328)	12,755
<i>Parliamentary behaviour</i>			
Debate Headings	+0.0031	(0.0105)	4,249
Core legal share	-0.0045	(0.0090)	4,249
Topic Model	+0.0112	(0.0077)	3,923
Wordfish (θ_z)	-0.0385	(0.0703)	3,917

Notes: Each row is a separate feols regression of the outcome on `union_sponsor` with MP and election-year fixed effects. Standard errors clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E6 — Sponsorship effect by MP working-class background



Notes: Same specification as the main sponsorship figure, estimated separately for working-class MPs (red triangles, N = 184 MPs), non-working-class MPs (blue squares, N = 841 MPs), and the full sample (grey circles). Standard errors clustered by MP.

E7 — Sponsorship event study: who is compared with whom

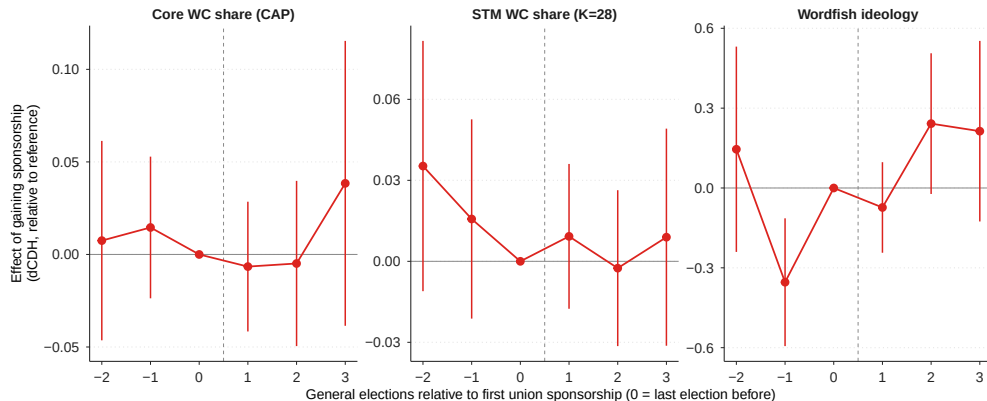
Event: F_i = first general election at which candidate i , *unsponsored at their first observed candidacy*, stands as union-sponsored. Event time $k = \text{GE} - (F_i - 1)$, so $k = 0$ is the last election before the gain (reference). Unit: candidate $\times$ GE term, Labour candidates with speech in ≥ 2 terms.

Sponsorship path (by term)	Candidates	Terms	Role in event study
Never sponsored (0 . . . 0)	634	2,438	Control at every horizon
Gains, keeps (0 . . . 0 1 . . . 1)	71	341	Switcher; not-yet control before F_i
Gains, later loses (0 . . . 1 . . . 0)	46	282	Switcher; not-yet control before F_i
Always sponsored (1 . . . 1)	214	890	Excluded: never gains
Sponsored at first term (1 . . . 0 . . .)	60	298	Excluded: no unsponsored baseline

Horizon	Switchers	Controls (candidates)	
		Never sponsored	Not yet sponsored
$\ell = 1$	91	332	40
$\ell = 2$	64	243	21
$\ell = 3$	51	186	11

- Effect ℓ : change in Y from $F_i - 1$ to $F_i - 1 + \ell$ for switchers vs. candidates observed at both dates and *still unsponsored* at $F_i - 1 + \ell$: never- plus not-yet-sponsored
- Already-sponsored terms are never controls (no “forbidden” comparisons)
- Gainers who later lose sponsorship remain switchers: the estimand is the effect of *having gained* sponsorship at F_i
- Excluded groups still enter the within-MP coefficient (Rung 4): the 1 . . . 0 candidates identify τ through their loss of sponsorship

E7b — Sponsorship event study: de Chaisemartin–D'Haultfoeuille

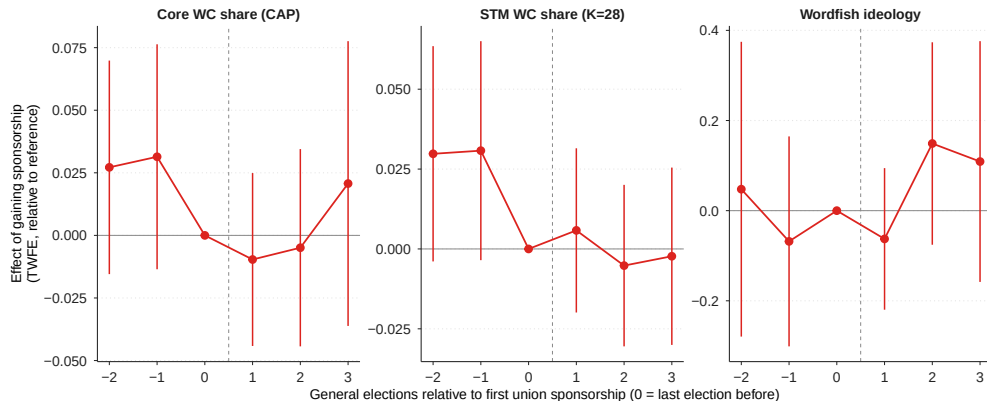


Notes: de Chaisemartin and D'Haultfoeuille (2026) dynamic estimator (`did_multipligt_dyn`, `switchers in`), on never-sponsored candidates plus candidates who gain sponsorship (comparison cells on the previous slide). $k = 0$ is the last general election before the first sponsored candidacy; $k < 0$ are placebos. Switchers at $k = 1, 2, 3$: 91, 64, 51 (CAP). Joint placebo p : CAP 0.74, STM 0.28,

Wordfish 0.01 (Wordfish is -0.35 at $k = -1$, before the gain). SEs clustered by candidate.

Two-way FE ES

E7c — Sponsorship event study: two-way fixed effects



Notes: Same sample and event-time axis as the dCdH figure. Event-time dummies for $k \in \{-2, -1, 1, 2, 3\}$, $k = 0$ omitted; switchers' terms outside $[-2, 3]$ dropped; never-sponsored candidates (all dummies zero) are the controls. Candidate and general-election fixed effects; SEs clustered by candidate. Exposed to the negative-weighting critique of staggered, non-absorbing designs; the heterogeneity-robust counterpart is the dCdH figure. Pre-gain coefficients at $k = -1$: CAP +0.031 (SE 0.023), STM +0.031 (SE 0.017).

F1 — Topic allocation and ideology: full table

	Debate Headings				Topic Model				Wordfish Ideology (θ_z)				Topic-Specific Wordfish			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Working-class	0.073*** (0.006)	0.049*** (0.006)	0.046*** (0.006)	0.046*** (0.006)	0.113*** (0.008)	0.085*** (0.008)	0.078*** (0.008)	0.077*** (0.008)	-0.144*** (0.048)	-0.021 (0.052)	-0.076 (0.053)	-0.076 (0.053)	-0.080** (0.032)	0.009 (0.034)	-0.029 (0.035)	-0.031 (0.034)
Female			0.015*** (0.005)	0.011** (0.005)			0.011** (0.004)	0.001 (0.004)			-0.268*** (0.047)	-0.161*** (0.045)			-0.090** (0.040)	-0.096** (0.040)
University			-0.008*** (0.002)	-0.008*** (0.002)			-0.016*** (0.002)	-0.016*** (0.002)			-0.073*** (0.023)	-0.075*** (0.023)			-0.069*** (0.019)	-0.068*** (0.019)
Private school			-0.004* (0.002)	-0.004* (0.002)			-0.003 (0.003)	-0.002 (0.003)			-0.066 (0.049)	-0.067 (0.049)			-0.080** (0.040)	-0.077* (0.040)
Aristocratic			-0.005** (0.002)	-0.003 (0.002)			-0.010*** (0.002)	-0.007*** (0.002)			-0.036 (0.032)	-0.036 (0.032)			-0.053* (0.030)	-0.052* (0.030)
Standard-Errors								Clustered by MP								
Observations	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	29,759	29,759	29,759	29,759
R ²	0.01792	0.10313	0.10493	0.10870	0.04568	0.12125	0.12707	0.14553	0.00097	0.07979	0.08423	0.10876	0.00042	0.08507	0.08693	0.09010
Party fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
Year fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
Decade cohort FEs				✓				✓				✓				✓

Notes: Columns 1–4 CAP core share; 5–8 STM $K=28$ core share; 9–12 Wordfish θ_z ; 13–16 topic-specific Wordfish within the Labour & Employment (CAP 5) and Housing (CAP 14) areas. Common in-service MP-year sample (51,680; 29,759 with a topic-specific score); party fixed effects for the seat party group. Standard errors clustered by MP.

F2 — Heterogeneity: full table

Table 5: Heterogeneity in the WC topic attention gap: STM and CAP outcomes

	Base	Seniority	Sponsored	Safe seat	Incumbent	Max'd
<i>Panel A: Topic Model</i>						
Working-class	0.0777*** (0.0082)	0.0779*** (0.0082) -0.0011 (0.0008)	0.0189 (0.0168)	0.0412*** (0.0101)	0.0564*** (0.0099)	0.0785*** (0.0105)
WC × Seniority						
WC × Union sponsored			0.0658*** (0.0189)			
WC × Safe seat				0.0410*** (0.0130)		
WC × Incumbent					0.0218** (0.0110)	
WC × Spending maxed						-0.0088 (0.0219)
University						
Private school						
Obs	51,680	51,680	51,680	38,516	36,184	16,698
R ²	0.127	0.136	0.130	0.115	0.116	0.118
FES	P+Y	P+Y	P+Y	P+Y	P+Y	P+Y
<i>Panel B: Debate Headings</i>						
Working-class	0.0460*** (0.0062)	0.0461*** (0.0061) -0.0008 (0.0005)	0.0218* (0.0121)	0.0239*** (0.0089)	0.0408*** (0.0099)	0.0463*** (0.0083)
WC × Seniority						
WC × Union sponsored			0.0250* (0.0139)			
WC × Safe seat				0.0248** (0.0112)		
WC × Incumbent					0.0022 (0.0109)	
WC × Spending maxed						-0.0157 (0.0164)
University						
Private school						
Obs	51,680	51,680	51,680	38,516	36,184	16,698
R ²	0.105	0.108	0.106	0.094	0.082	0.085
FES	P+Y	P+Y	P+Y	P+Y	P+Y	P+Y

Each column is a separate OLS regression with seat-party + year fixed effects (P+Y) and controls (female, university, private school, aristocratic connections); SEs clustered by MP. Common MP-year sample of Table ??; each moderator column restricts to MP-years where the moderator is observed. "Base" has no moderator; the remaining columns add Working-class × moderator.

Topic Model (STM, K=28): topics 11 (Labour/unemployment), 21 (Trade unions), 18 (Mining/coal). Debate Headings (CAP): shares of CAP major topics 5 (Labour/Employment), 13 (Social Welfare).

Seniority = year – first election year, centred at 10. Safe seat = margin ≥ 10% at the last election. Incumbent = sitting MP at the election. Spending maxed = constituency-candidate at the legal spending limit in every observed election (Fournales).

*p < 0.10, **p < 0.05, ***p < 0.01.

F3 — By historical period: full table

	Pre-Ref.		Vict.-Ed.		Interwar		Post-War		Thatcher		New Lab.	
	All	Lab	All	Lab	All	Lab	All	Lab	All	Lab	All	Lab
Working-class			0.0354 (0.0277)	0.0345 (0.0319)	0.0764*** (0.0163)	0.0683*** (0.0180)	0.0712*** (0.0127)	0.0676*** (0.0131)	0.1005*** (0.0231)	0.1049*** (0.0252)	0.0318** (0.0150)	0.0354** (0.0159)
Obs	4,933		11,229	445	6,147	1,849	11,922	5,876	7,648	3,343	9,801	5,205
R ²	0.043		0.183	0.105	0.135	0.128	0.083	0.099	0.084	0.099	0.084	0.055
FES	P+Y	Y	P+Y	Y	P+Y	Y	P+Y	Y	P+Y	Y	P+Y	Y

Notes: Outcome is the STM $K=28$ core WC topic share (topics 11+21+18). All = seat-party + year FE; Lab = MP-years in Labour-held seats, year FE. Controls: female, university, private school, aristocratic connections. No working-class MP sits before 1868, so the Pre-Reform coefficient is not estimable. SEs clustered by MP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F4 — Language complexity: full table

	TTR (lexical diversity)				Syllables per word				Concreteness			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Working-class	-0.030 (0.035)	-0.104*** (0.027)	-0.095*** (0.027)	-0.096*** (0.027)	-0.348*** (0.065)	-0.297*** (0.066)	-0.253*** (0.066)	-0.253*** (0.064)	0.532*** (0.051)	0.275*** (0.047)	0.173*** (0.048)	0.171*** (0.048)
Female			0.076*** (0.025)	0.045* (0.025)			0.055 (0.044)	-0.000 (0.043)			0.189*** (0.037)	0.148*** (0.036)
University			0.005 (0.012)	0.009 (0.012)			0.091*** (0.022)	0.104*** (0.022)			-0.252*** (0.019)	-0.255*** (0.019)
Private school			-0.032 (0.028)	-0.026 (0.028)			0.010 (0.048)	0.019 (0.046)			0.046 (0.039)	0.044 (0.040)
Aristocratic			0.029 (0.020)	0.026 (0.020)			-0.023 (0.028)	0.008 (0.028)			-0.073*** (0.026)	-0.056** (0.026)
Standard-Errors	Clustered by MP											
Observations	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680	51,680
R ²	0.00004	0.20539	0.20578	0.20907	0.00564	0.24272	0.24471	0.25829	0.01318	0.38564	0.40292	0.40795
Raw mean of outcome			0.632			1.525				2.329		
Raw SD of outcome			0.040			0.053				0.050		
Party fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Year fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Individual controls			✓	✓			✓	✓			✓	✓
Decade cohort FEs				✓				✓				✓

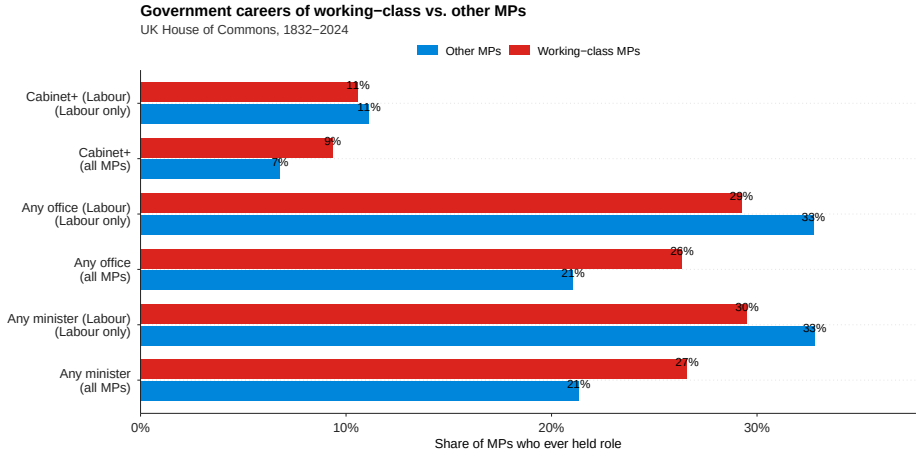
Notes: Outcomes standardised to unit variance. For each outcome: bare OLS; seat-party and year fixed effects; the paper-wide individual controls; and decade-of-first-election cohort fixed effects. Common in-service MP-year sample. Standard errors clustered by MP.

F5 — Party loyalty and government careers: full table

	Cohesion				Any office				Cabinet+			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Working-class	0.073*** (0.004)	0.011 (0.015)	0.015 (0.017)	0.012 (0.017)	0.040* (0.021)	-0.084*** (0.023)	-0.081*** (0.022)	-0.025 (0.022)	0.021 (0.013)	-0.024* (0.014)	-0.024 (0.015)	0.002 (0.015)
University			0.010** (0.004)	0.009** (0.004)				0.147*** (0.009)				0.062*** (0.006)
Private school			0.005 (0.006)	0.004 (0.006)				-0.025* (0.014)				-0.007 (0.009)
Aristocratic			0.014*** (0.004)	0.014*** (0.004)				0.120*** (0.012)				0.067*** (0.008)
Female								-0.046* (0.025)				0.010 (0.016)
Standard-Errors	wid				IID							
Observations	9,518	9,518	9,518	9,518	9,279	9,279	9,279	9,279	9,279	9,279	9,279	9,279
R ²	0.00028	0.18959	0.19258	0.19418	0.00037	0.04025	0.10971	0.15026	0.00027	0.01201	0.02552	0.04751
Party fixed effects		✓	✓	✓		✓	✓	✓		✓	✓	✓
Parliament fixed effects		✓	✓	✓								
Decade cohort fixed effects				✓			✓	✓			✓	✓

Notes: Cohesion is the share of divisions in which the MP voted with the majority of their own party (Eggers-Spirling, 1832–1910). Any office = 1 if the MP ever held office above backbencher; Cabinet+ = 1 if the MP ever reached Cabinet. Party fixed effects: the MP's modal seat party group; cohort: decade of first recorded election.

F6 — Government careers of working-class vs other MPs



References I

- Abbott, J. and Deveau, F. (2024). Do working-class candidates activate class-based voting? *Electoral Studies*, 92:102862.
- Abbott, J. and DeVeaux, F. (2026). What Happens When Working-Class Candidates Run for Office? Working paper.
- Acemoglu, D., Naidu, S., Restrepo, P., and Robinson, J. A. (2015). Democracy, Redistribution, and Inequality. In Atkinson, A. B. and Bourguignon, F., editors, *Handbook of Income Distribution*, volume 2, pages 1885–1966. Elsevier, Amsterdam.
- Acemoglu, D. and Robinson, J. A. (2006). *Economic Origins of Dictatorship and Democracy*. Cambridge University Press, Cambridge.
- Acemoglu, D. and Robinson, J. A. (2008). Persistence of power, elites, and institutions. *American Economic Review*, 98(1):267–293.
- Barnes, T. D. and Saxton, G. W. (2019). Working-Class Legislators and Perceptions of Representation in Latin America. *Political Research Quarterly*, 72(4):910–928.
- Berlinski, S., Dewan, T., and Van Coppenolle, B. (2014). Franchise extension and the british aristocracy. *Legislative Studies Quarterly*, 39(4):531–558.
- Besley, T. and Coate, S. (1997). An economic model of representative democracy. *The Quarterly Journal of Economics*, 112(1):85–114.

References II

- Besley, T., Dunn, C. M., Geiecke, F. C., and Larcinese, V. (2023). Aristocratic Networks in British Politics: 1831 to the Present. Working paper, London School of Economics.
- Brysbaert, M., Warriner, A. B., and Kuperman, V. (2014). Concreteness Ratings for 40 Thousand Generally Known English Word Lemmas. *Behavior Research Methods*, 46(3):904–911.
- Cagé, J. and Dewitte, E. (2025). It Takes Money to Make MPs: New Evidence from 150 Years of British Campaign Spending. *The Journal of Economic History*.
- Carnes, N. (2012). Does the numerical underrepresentation of the working class in Congress matter? *Legislative Studies Quarterly*, 37(1):5–34.
- Carnes, N. (2013). *White-Collar Government: The Hidden Role of Class in Economic Policy Making*. University of Chicago Press, Chicago.
- Carnes, N., Ferrer, J., Golden, M., Lillywhite, E., Lupu, N., and Nazrullaeva, E. (2025). The Global Legislators Database: Characteristics of National Legislators in the World's Democracies. *British Journal of Political Science*, 55:e27.
- Carnes, N. and Lupu, N. (2016). Do voters dislike working-class candidates? Voter biases and the descriptive underrepresentation of the working class. *American Political Science Review*, 110(4):832–844.
- Carnes, N. and Lupu, N. (2023a). Are there social class gaps in nascent political ambition? Survey evidence from the Americas.

References III

- Carnes, N. and Lupu, N. (2023b). The economic backgrounds of politicians. *Annual Review of Political Science*, 26(1):253–270.
- Chattopadhyay, R. and Duflo, E. (2004). Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*, 72(5):1409–1443.
- Corvalan, A., Querubin, P., and Vicente, S. (2020). The political class and redistributive policies. *Journal of the European Economic Association*, 18(1):1–48.
- Costa, M. and Pereira, M. M. (2026). Why parties can benefit from promoting occupational diversity in legislatures: Experimental evidence from three countries. *American Journal of Political Science*, 70(1):304–320.
- Dal Bó, E., Finan, F., Folke, O., Persson, T., and Rickne, J. (2017). Who becomes a politician? *The Quarterly Journal of Economics*, 132(4):1877–1914.
- de Chaisemartin, C. and D'Haultfœuille, X. (2026). Difference-in-Differences Estimators of Intertemporal Treatment Effects. *Review of Economics and Statistics*, 108(4):863–880.
- Eggers, A. C. and Hainmueller, J. (2009). MPs for sale? Returns to office in postwar British politics. *American Political Science Review*, 103(4):513–533.
- Elsässer, L. and Schäfer, A. (2023). Political inequality in rich democracies. *Annual Review of Political Science*, 26(1):469–487.

References IV

- Evans, G. and Tilley, J. (2017). *The new politics of class: The political exclusion of the British working class*. Oxford University Press.
- Folke, O. and Rickne, J. (2024). The Class Ceiling in Politics. *American Political Science Review*, pages 1–18.
- Fourinaies, A. (2021). How do campaign spending limits affect elections? Evidence from the United Kingdom 1885–2019. *American Political Science Review*, 115(2):395–411.
- Fourinaies, A. (2024). Can interest groups influence elections? Evidence from unions in Great Britain, 1900–2019. *American Journal of Political Science*.
- Fujiwara, T. (2015). Voting Technology, Political Responsiveness, and Infant Health: Evidence from Brazil. *Econometrica*, 83(2):423–464.
- Gilens, M. (2012). Affluence and influence: Economic inequality and political power in America. In *Affluence and influence*. Princeton University Press.
- Göbel, S. and Munzert, S. (2022). The comparative legislators database. *British Journal of Political Science*, 52(3):1398–1408.
- Goet, N. D. (2019). Measuring polarization with text analysis: Evidence from the UK House of Commons, 1811–2015. *Political Analysis*, 27(4):518–539.
- Heath, O. (2015). Policy Representation, Social Representation and Class Voting in Britain. *British Journal of Political Science*, 45(1):173–193.

References V

- Korpi, W. (1983). *The Democratic Class Struggle*. Routledge and Kegan Paul, London.
- Langsæther, P. E. and Nyrup, J. (2026). Which governments are most representative? *Available at SSRN* 7391238.
- Larcinese, V. (2024). Enfranchisement and representation: evidence from the introduction of “quasi-universal” suffrage in Italy. *The Journal of Politics*, 86(2):565–581.
- Lupu, N. and Pontusson, J. (2023). *Unequal democracies: public policy, responsiveness, and redistribution in an era of rising economic inequality*. Cambridge University Press.
- Lupu, N. and Warner, Z. (2022). Affluence and congruence: unequal representation around the world. *The Journal of Politics*, 84(1):276–290.
- Marshall, J. (2024). Can close election regression discontinuity designs identify effects of winning politician characteristics? *American Journal of Political Science*, 68(2):494–510.
- Meltzer, A. H. and Richard, S. F. (1981). A Rational Theory of the Size of Government. *Journal of Political Economy*, 89(5):914–927.
- Mor, M. and Boix, C. (2024). Social Democracy and the Birth of Working-Class Representation in Europe. *World Politics*, 76(3):499–542.
- Mosca, G. (1939). *The Ruling Class*. McGraw-Hill, New York. Original: *Elementi di Scienza Politica*, 1896.

References VI

- O'Grady, T. (2019). Careerists versus coal-miners: welfare reforms and the substantive representation of social groups in the British Labour Party. *Comparative Political Studies*, 52(4):544–578.
- Pande, R. (2003). Can Mandated Political Representation Increase Policy Influence for Disadvantaged Minorities? Theory and Evidence from India. *American Economic Review*, 93(4):1132–1151.
- Pareto, V. (1935). *The Mind and Society*. Harcourt, Brace, New York. Original: *Trattato di Sociologia Generale*, 1916.
- Piketty, T. and Cagé, J. (2023). *Une histoire du conflit politique: Elections et inégalités sociales en France, 1789-2022*. Seuil.
- Poertner, M. (2023). Does political representation increase participation? Evidence from party candidate lotteries in Mexico. *American Political Science Review*, 117(2):537–556.
- Praninskas, G. and Munroe, E. (2025). Wealth and Intergenerational Political Power Mobility. Working Paper.
- Proksch, S.-O. and Slapin, J. B. (2012). Institutional foundations of legislative speech. *American Journal of Political Science*, 56(3):520–537.
- Przeworski, A. and Sprague, J. (1986). *Paper Stones: A History of Electoral Socialism*. University of Chicago Press, Chicago.
- Roberts, M. E., Stewart, B. M., Tingley, D., Airolidi, E. M., et al. (2013). The structural topic model and applied social science. In *Advances in neural information processing systems workshop on topic models: computation, application, and evaluation*, volume 4, pages 1–20. Harrahs and Harveys, Lake Tahoe.

References VII

- Sassoon, D. (1996). *One Hundred Years of Socialism: The West European Left in the Twentieth Century*. I.B. Tauris, London.
- Slapin, J. B. and Proksch, S.-O. (2008). A scaling model for estimating time-series party positions from texts. *American Journal of Political Science*, 52(3):705–722.
- Wauters, B. (2012). Blue collars striking the red flag: formal and descriptive representation of the working class in the Belgian House of Representatives 1946–2007. *Labor History*, 53(2):225–243.

- Abbott, J. and Deveau, F. (2024). Do working-class candidates activate class-based voting? *Electoral Studies*, 92:102862.
- Abbott, J. and DeVeaux, F. (2026). What Happens When Working-Class Candidates Run for Office? Working paper.
- Acemoglu, D., Naidu, S., Restrepo, P., and Robinson, J. A. (2015). Democracy, Redistribution, and Inequality. In Atkinson, A. B. and Bourguignon, F., editors, *Handbook of Income Distribution*, volume 2, pages 1885–1966. Elsevier, Amsterdam.
- Acemoglu, D. and Robinson, J. A. (2006). *Economic Origins of Dictatorship and Democracy*. Cambridge University Press, Cambridge.
- Acemoglu, D. and Robinson, J. A. (2008). Persistence of power, elites, and institutions. *American Economic Review*, 98(1):267–293.
- Barnes, T. D. and Saxton, G. W. (2019). Working-Class Legislators and Perceptions of Representation in Latin America. *Political Research Quarterly*, 72(4):910–928.
- Berlinski, S., Dewan, T., and Van Coppenolle, B. (2014). Franchise extension and the British aristocracy. *Legislative Studies Quarterly*, 39(4):531–558.
- Besley, T. and Coate, S. (1997). An economic model of representative democracy. *The Quarterly Journal of Economics*, 112(1):85–114.

- Besley, T., Dunn, C. M., Geiecke, F. C., and Larcinese, V. (2023). Aristocratic Networks in British Politics: 1831 to the Present. Working paper, London School of Economics.
- Brysbaert, M., Warriner, A. B., and Kuperman, V. (2014). Concreteness Ratings for 40 Thousand Generally Known English Word Lemmas. *Behavior Research Methods*, 46(3):904–911.
- Cagé, J. and Dewitte, E. (2025). It Takes Money to Make MPs: New Evidence from 150 Years of British Campaign Spending. *The Journal of Economic History*.
- Carnes, N. (2012). Does the numerical underrepresentation of the working class in Congress matter? *Legislative Studies Quarterly*, 37(1):5–34.
- Carnes, N. (2013). *White-Collar Government: The Hidden Role of Class in Economic Policy Making*. University of Chicago Press, Chicago.
- Carnes, N., Ferrer, J., Golden, M., Lillywhite, E., Lupu, N., and Nazrullaeva, E. (2025). The Global Legislators Database: Characteristics of National Legislators in the World's Democracies. *British Journal of Political Science*, 55:e27.
- Carnes, N. and Lupu, N. (2016). Do voters dislike working-class candidates? Voter biases and the descriptive underrepresentation of the working class. *American Political Science Review*, 110(4):832–844.
- Carnes, N. and Lupu, N. (2023a). Are there social class gaps in nascent political ambition? Survey evidence from the Americas.

- Carnes, N. and Lupu, N. (2023b). The economic backgrounds of politicians. *Annual Review of Political Science*, 26(1):253–270.
- Chattopadhyay, R. and Duflo, E. (2004). Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*, 72(5):1409–1443.
- Corvalan, A., Querubin, P., and Vicente, S. (2020). The political class and redistributive policies. *Journal of the European Economic Association*, 18(1):1–48.
- Costa, M. and Pereira, M. M. (2026). Why parties can benefit from promoting occupational diversity in legislatures: Experimental evidence from three countries. *American Journal of Political Science*, 70(1):304–320.
- Dal Bó, E., Finan, F., Folke, O., Persson, T., and Rickne, J. (2017). Who becomes a politician? *The Quarterly Journal of Economics*, 132(4):1877–1914.
- de Chaisemartin, C. and D'Haultfœuille, X. (2026). Difference-in-Differences Estimators of Intertemporal Treatment Effects. *Review of Economics and Statistics*, 108(4):863–880.
- Eggers, A. C. and Hainmueller, J. (2009). MPs for sale? Returns to office in postwar British politics. *American Political Science Review*, 103(4):513–533.
- Elsässer, L. and Schäfer, A. (2023). Political inequality in rich democracies. *Annual Review of Political Science*, 26(1):469–487.

- Evans, G. and Tilley, J. (2017). *The new politics of class: The political exclusion of the British working class*. Oxford University Press.
- Folke, O. and Rickne, J. (2024). The Class Ceiling in Politics. *American Political Science Review*, pages 1–18.
- Fourinaies, A. (2021). How do campaign spending limits affect elections? Evidence from the United Kingdom 1885–2019. *American Political Science Review*, 115(2):395–411.
- Fourinaies, A. (2024). Can interest groups influence elections? Evidence from unions in Great Britain, 1900–2019. *American Journal of Political Science*.
- Fujiwara, T. (2015). Voting Technology, Political Responsiveness, and Infant Health: Evidence from Brazil. *Econometrica*, 83(2):423–464.
- Gilens, M. (2012). Affluence and influence: Economic inequality and political power in America. In *Affluence and influence*. Princeton University Press.
- Göbel, S. and Munzert, S. (2022). The comparative legislators database. *British Journal of Political Science*, 52(3):1398–1408.
- Goet, N. D. (2019). Measuring polarization with text analysis: Evidence from the UK House of Commons, 1811–2015. *Political Analysis*, 27(4):518–539.
- Heath, O. (2015). Policy Representation, Social Representation and Class Voting in Britain. *British Journal of Political Science*, 45(1):173–193.

- Korpi, W. (1983). *The Democratic Class Struggle*. Routledge and Kegan Paul, London.
- Langsæther, P. E. and Nyrup, J. (2026). Which governments are most representative? Available at SSRN 7391238.
- Larcinese, V. (2024). Enfranchisement and representation: evidence from the introduction of “quasi-universal” suffrage in Italy. *The Journal of Politics*, 86(2):565–581.
- Lupu, N. and Pontusson, J. (2023). *Unequal democracies: public policy, responsiveness, and redistribution in an era of rising economic inequality*. Cambridge University Press.
- Lupu, N. and Warner, Z. (2022). Affluence and congruence: unequal representation around the world. *The Journal of Politics*, 84(1):276–290.
- Marshall, J. (2024). Can close election regression discontinuity designs identify effects of winning politician characteristics? *American Journal of Political Science*, 68(2):494–510.
- Meltzer, A. H. and Richard, S. F. (1981). A Rational Theory of the Size of Government. *Journal of Political Economy*, 89(5):914–927.
- Mor, M. and Boix, C. (2024). Social Democracy and the Birth of Working-Class Representation in Europe. *World Politics*, 76(3):499–542.
- Mosca, G. (1939). *The Ruling Class*. McGraw-Hill, New York. Original: *Elementi di Scienza Politica*, 1896.

- O'Grady, T. (2019). Careerists versus coal-miners: welfare reforms and the substantive representation of social groups in the British Labour Party. *Comparative Political Studies*, 52(4):544–578.
- Pande, R. (2003). Can Mandated Political Representation Increase Policy Influence for Disadvantaged Minorities? Theory and Evidence from India. *American Economic Review*, 93(4):1132–1151.
- Pareto, V. (1935). *The Mind and Society*. Harcourt, Brace, New York. Original: *Trattato di Sociologia Generale*, 1916.
- Piketty, T. and Cagé, J. (2023). *Une histoire du conflit politique: Elections et inégalités sociales en France, 1789-2022*. Seuil.
- Poertner, M. (2023). Does political representation increase participation? Evidence from party candidate lotteries in Mexico. *American Political Science Review*, 117(2):537–556.
- Praninskas, G. and Munroe, E. (2025). Wealth and Intergenerational Political Power Mobility. Working Paper.
- Proksch, S.-O. and Slapin, J. B. (2012). Institutional foundations of legislative speech. *American Journal of Political Science*, 56(3):520–537.
- Przeworski, A. and Sprague, J. (1986). *Paper Stones: A History of Electoral Socialism*. University of Chicago Press, Chicago.

- Roberts, M. E., Stewart, B. M., Tingley, D., Airoldi, E. M., et al. (2013). The structural topic model and applied social science. In *Advances in neural information processing systems workshop on topic models: computation, application, and evaluation*, volume 4, pages 1–20. Harrahs and Harveys, Lake Tahoe.
- Sassoon, D. (1996). *One Hundred Years of Socialism: The West European Left in the Twentieth Century*. I.B. Tauris, London.
- Slapin, J. B. and Proksch, S.-O. (2008). A scaling model for estimating time-series party positions from texts. *American Journal of Political Science*, 52(3):705–722.
- Wauters, B. (2012). Blue collars striking the red flag: formal and descriptive representation of the working class in the Belgian House of Representatives 1946–2007. *Labor History*, 53(2):225–243.